ESRG

Embedded Systems Research Group

CENTROALGORITMI

Secure Asset Guard

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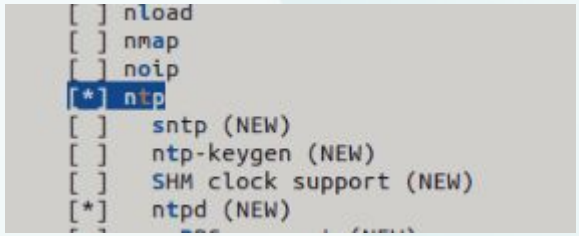
COMPETE

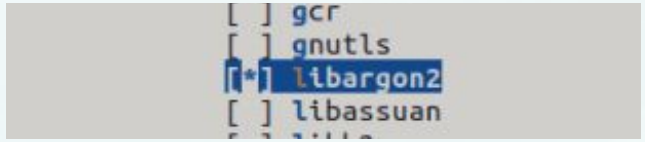
QREN Quadro de Referência Estratégico Nacional

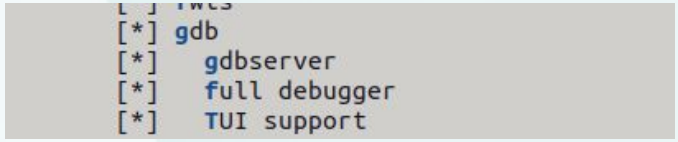
União Europeia
Fundo Europeu de Desenvolvimento Regional

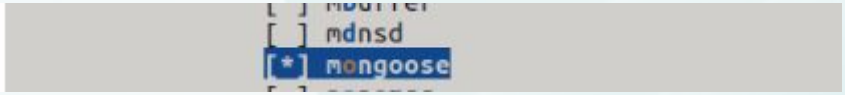
FCT Fundação para a Ciência e a Tecnologia
Ministério da Ciência, Tecnologia e Ensino Superior

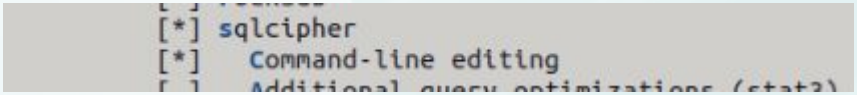
Buildroot libraries

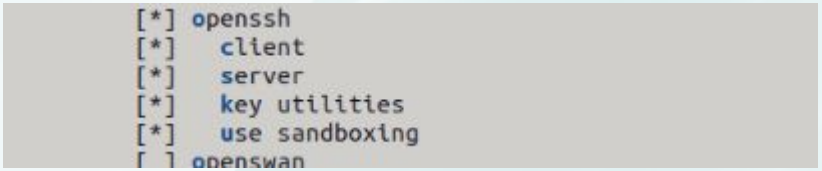




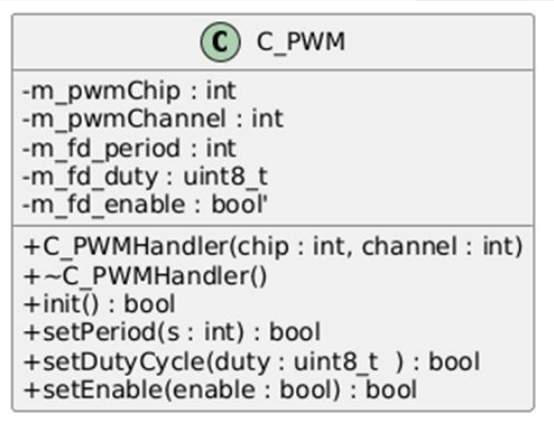








PWM




```cpp
class C_PWM
{
private:
    int m_pwmChip;
    int m_pwmChannel;
    int m_fd_period;
    uint8_t m_fd_duty;
    bool m_fd_enable;
public:
    // Constructor
    C_PWM(int chip, int channel);
    // Destructor
    ~C_PWM();
    // inicialization
    bool init();
    // period
    bool setPeriodns(int s);
    //duty cyle percentage
    bool setDutyCycle(uint8_t duty);
    // activates/deactivates pwm
    bool setEnable(bool enable);
};
```

Device tree PWM overlay


```
# cd /sys/class/pwm/
# ls
#
```

dtoverlay=pwm,pin=12,func=4

```
# cd /sys/class/pwm/
# ls
pwmchip0
#
```

BCM2711 Datasheet

Bits	Name	Description	Type	Reset
31:30	Reserved.	-	-	-
29:27	FSEL9	FSEL9 - Function Select 9 000 = GPIO Pin 9 is an input 001 = GPIO Pin 9 is an output 100 = GPIO Pin 9 takes alternate function 0 101 = GPIO Pin 9 takes alternate function 1 110 = GPIO Pin 9 takes alternate function 2 111 = GPIO Pin 9 takes alternate function 3 011 = GPIO Pin 9 takes alternate function 4 010 = GPIO Pin 9 takes alternate function 5	RW	0x0
26:24	FSEL8	FSEL8 - Function Select 8	RW	0x0
23:21	FSEL7	FSEL7 - Function Select 7	RW	0x0
20:18	FSEL6	FSEL6 - Function Select 6	RW	0x0
17:15	FSEL5	FSEL5 - Function Select 5	RW	0x0
14:12	FSEL4	FSEL4 - Function Select 4	RW	0x0
11:9	FSEL3	FSEL3 - Function Select 3	RW	0x0
8:6	FSEL2	FSEL2 - Function Select 2	RW	0x0
5:3	FSEL1	FSEL1 - Function Select 1	RW	0x0
2:0	FSEL0	FSEL0 - Function Select 0	RW	0x0

5.1.2 GPIO Alternate Functions

GPIO	Default Pull	ALT0	ALT1	ALT2	ALT3	ALT4	ALT5
0	High	SDA0	SA5	PCLK	SPI3_CE0_N	TXD2	SDA6
1	High	SCL0	SA4	DE	SPI3_MISO	RXD2	SCL6
2	High	SDA1	SA3	LCD_VSYNC	SPI3_MOSI	CTS2	SDA3
3	High	SCL1	SA2	LCD_HSYNC	SPI3_SCLK	RTS2	SCL3
4	High	GPCLK0	SA1	DPI_D0	SPI4_CE0_N	TXD3	SDA3
5	High	GPCLK1	SA0	DPI_D1	SPI4_MISO	RXD3	SCL3
6	High	GPCLK2	SOE_N	DPI_D2	SPI4_MOSI	CTS3	SDA4
7	High	SPI0_CE1_N	SWE_N	DPI_D3	SPI4_SCLK	RTS3	SCL4
8	High	SPI0_CE0_N	SD0	DPI_D4	-	TXD4	SDA4
9	Low	SPI0_MISO	SD1	DPI_D5	-	RXD4	SCL4
10	Low	SPI0_MOSI	SD2	DPI_D6	-	CTS4	SDA5
11	Low	SPI0_SCLK	SD3	DPI_D7	-	RTS4	SCL5
12	Low	PWM0	SD4	DPI_D8	SPI5_CE0_N	TXD5	SDA5
13	Low	PWM1	SD5	DPI_D9	SPI5_MISO	RXD5	SCL5

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```
# cd /sys/class/pwm/pwmchip0/
# ls
device      export      npwm        power       subsystem   uevent      unexport
# echo 0 > export
# ls
device      export      npwm        power       pwm0        subsystem   uevent      unexport
```

PWM Channel 0 Enabled

```
bool C_PWM::init()
{
    string exportPath = "/sys/class/pwm/pwmchip" + to_string(m_pwmChip)+ "/export";

    int fd = open( file: exportPath.c_str(), oflag: O_WRONLY);
    if (fd < 0)
    {
        cerr << "Erro ao abrir export\n";
        return false;
    }

    string channel = to_string(m_pwmChannel);
    if (write(fd, buf: channel.c_str(), n: channel.size()) < 0)
    {
        cerr << "Erro ao exportar PWM\n";
        close(fd);
        return false;
    }

    close(fd);
    return true;
}
```

```
int open(const char *pathname, int flags, mode_t mode);
```

- open() returns -1 if not successful and returns file descriptor if successful

```
ssize_t write(int fd, const void *buf, size_t count);
```

- write() returns -1 if not successful and returns number of bytes written if successful

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```
# cd /sys/class/pwm/pwmchip0/pwm0/
# ls
capture     duty_cycle  enable      period      polarity    power       uevent
```

Linux is expecting nanoseconds in Period and Duty_Cycle and 0 or 1 in Enable

```cpp
bool C_PWM::setDutyCycle(uint8_t duty) {
    m_fd_duty = duty;
    int dutyns = (m_fd_period * duty) / 100;

    // Caminho para o ficheiro duty_cycle
    string dutyPath =
            "/sys/class/pwm/pwmchip" + to_string(m_pwmChip) +
            "/pwm" + to_string(m_pwmChannel) + "/duty_cycle";
    int fd = open( file: dutyPath.c_str(),  oflag: O_WRONLY);
    if (fd < 0) {
        cerr << "Erro ao abrir duty_cycle\n";
        return false;
    }
    string val = to_string(dutyns);
    if (write(fd,  buf: val.c_str(),  n: val.size()) < 0) {
        cerr << "Erro ao escrever duty_cycle\n";
        close(fd);
        return false;
    }
    close(fd);
    return true;
}
```

```cpp
bool C_PWM::setPeriodns(int s) {
    m_fd_period = s;
    string periodPath =
            "/sys/class/pwm/pwmchip" + to_string(m_pwmChip) +
            "/pwm" + to_string(m_pwmChannel) + "/period";
    int fd = open( file: periodPath.c_str(),  oflag: O_WRONLY);
    if (fd < 0)
    {
        cerr << "Erro ao abrir period\n";
        return false;
    }
    string val = to_string(s);
    if (write(fd,  buf: val.c_str(),  n: val.size()) < 0) {
        cerr << "Erro ao escrever period\n";
        close(fd);
        return false;
    }
    close(fd);
    return true;
}
```

```cpp
bool C_PWM::setEnable(bool enable)
{
    m_fd_enable = enable;
    string enablePath =
            "/sys/class/pwm/pwmchip" + to_string(m_pwmChip) +
            "/pwm" + to_string(m_pwmChannel) + "/enable";

    int fd = open( file: enablePath.c_str(),  oflag: O_WRONLY);
    if (fd < 0)
    {
        cerr << "Erro ao abrir enable\n";
        return false;
    }
    string val = enable ? "1" : "0";
    if (write(fd,  buf: val.c_str(),  n: val.size()) < 0)
    {
        cerr << "Erro ao escrever enable\n";
        close(fd);
        return false;
    }
    close(fd);
    return true;
}
```

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Interface

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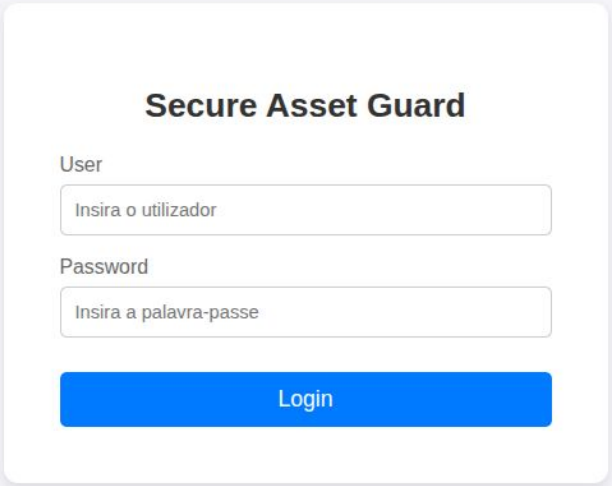
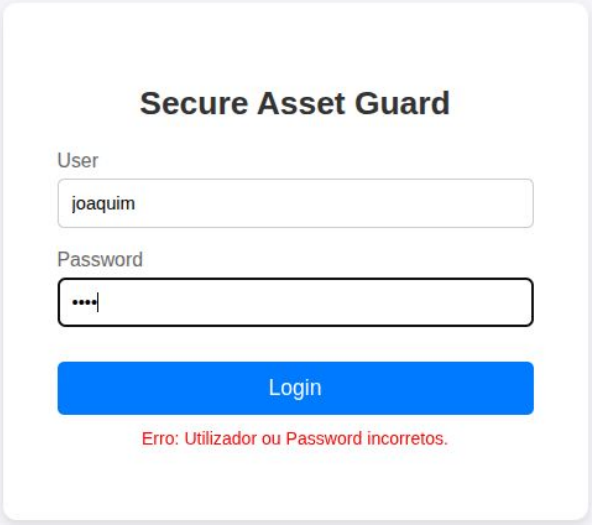
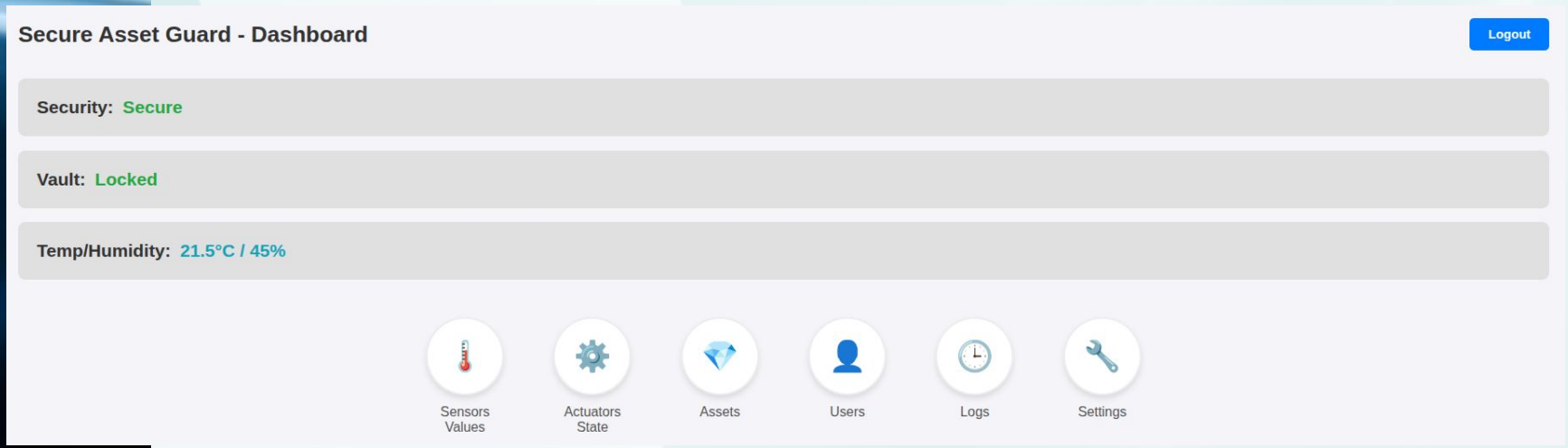






Device tree UART overlay


```
# cd /sys/class/tty
# ls
```

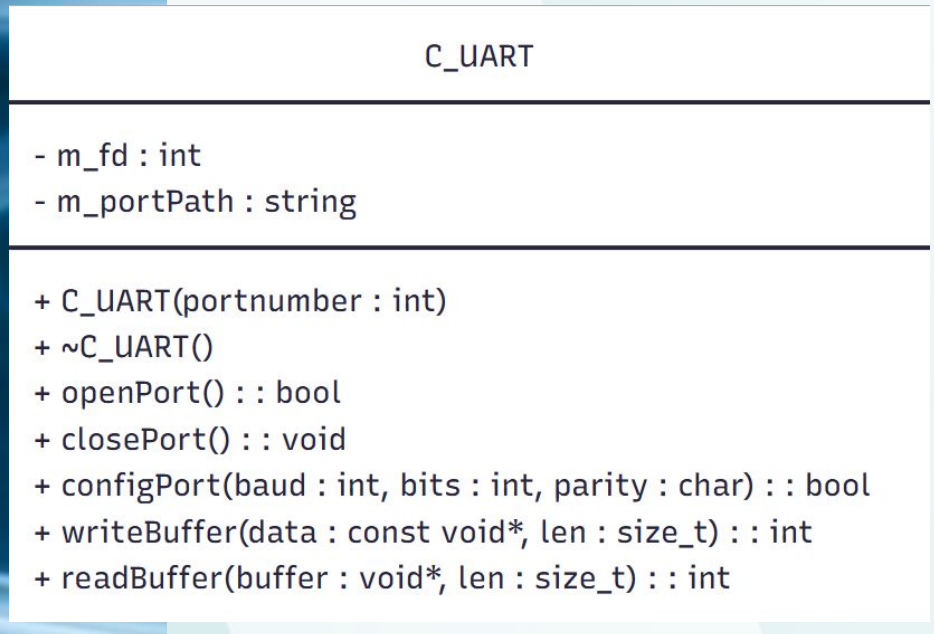
```
dtoverlay=uart2
```

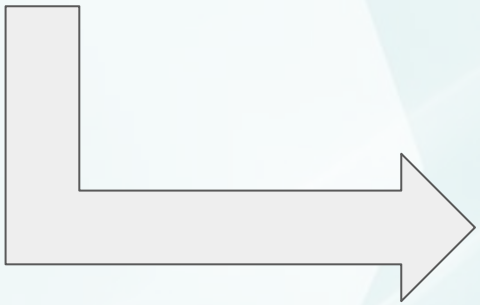
```
# cd /dev/
# ls
ttyAMA2
```

```
# cd /sys/class/tty
# ls
ttyAMA2
```

GPIO	Default Pull	ALT0	ALT1	ALT2	ALT3	ALT4	ALT5
0	High	SDA0	SA5	PCLK	SPI3_CE0_N	TXD2	SDA6
1	High	SCL0	SA4	DE	SPI3_MISO	RXD2	SCL6

Class UART





```
class C_UART {

    int m_fd;
    string m_portPath;
public:
    C_UART(int portnumber);

    // Destrutor
    ~C_UART();

    bool openPort();

    void closePort();

    bool configPort(int baud, int bits, char parity);

    int writeBuffer(const void* data, size_t len);

    int readBuffer(void* buffer, size_t len);

};
```

Device tree I2C overlay


```
# cd /sys/class/i2c-adapter
# ls
#
```

```
#i2c1 Sda and Scl mapped by predefinition to GPIO 2 and 3
dtoverlay=i2c1
```

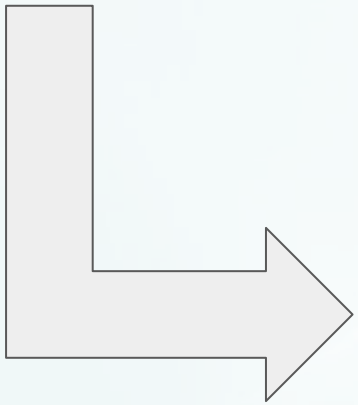
```
# cd /sys/class/i2c-adapter
# ls
i2c-1
```

```
# cd /dev/
# ls
autofs
bus
cachefiles
console
cpu_dma_latency
dma_heap
fd
full
gpiochip0
gpiochip1
hwrng
i2c-1
```

GPIO	Default Pull	ALT0	ALT1	ALT2	ALT3	ALT4	ALT5
2	High	SDA1	SA3	LCD_VSYNC	SPI3_MOSI	CTS2	SDA3
3	High	SCL1	SA2	LCD_HSYNC	SPI3_SCLK	RTS2	SCL3

Class I2C

C_I2C
- m_slaveaddress : uint8_t - m_fd : int - m_devicePath : string
+ C_I2C(i2cbusnum : int, slave_address : uint8_t) + ~C_I2C() + init() : : bool + closeI2C() : : void + writeRegister(reg : uint8_t, value : uint8_t) : : bool + readRegister(reg : uint8_t, value : uint8_t*) : : bool + readBytes(reg : uint8_t, buffer : uint8_t*, len : size_t) : : bool



```
class C_I2C {
    uint8_t m_slaveaddress;
    int m_fd;
    string m_devicePath;
public:
    C_I2C(int i2cbusnum, uint8_t slave_address);
    ~C_I2C();
    //open dev directory and "set" slave address
    bool init();
    //close fd
    void closeI2C();
    //used mainly to write some configuration in sensor specific register
    bool writeRegister(uint8_t reg, uint8_t value);
    //used to read only one sensor specific reg
    bool readRegister(uint8_t reg, uint8_t& value);
    //used to read more than one reg
    bool readBytes(uint8_t reg, uint8_t* buffer, size_t len);

};
```

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```
C_I2C::C_I2C(int i2cbusnum, uint8_t slave_address)
    : m_slaveaddress(slave_address), m_fd(-1)
{

    m_devicePath = "/dev/i2c-" + to_string(i2cbusnum);
}


C_I2C::~C_I2C() {
    closeI2C();
}

bool C_I2C::init() {
    m_fd = open( file: m_devicePath.c_str(), oflag: O_RDWR);
    if (m_fd < 0) {
        cerr << "C_I2C: Erro ao abrir dev/...\n";
        return false;
    }

    if (ioctl(m_fd, request: I2C_SLAVE, m_slaveaddress) < 0) {
        cerr<<"C_I2C: Erro ao definir slave (ioctl)\n";
        return false;
    }

    return true;
}
```

ioctl() to define the slave address that i2c driver will be using in the beginning of write() and read()

Class GPIO

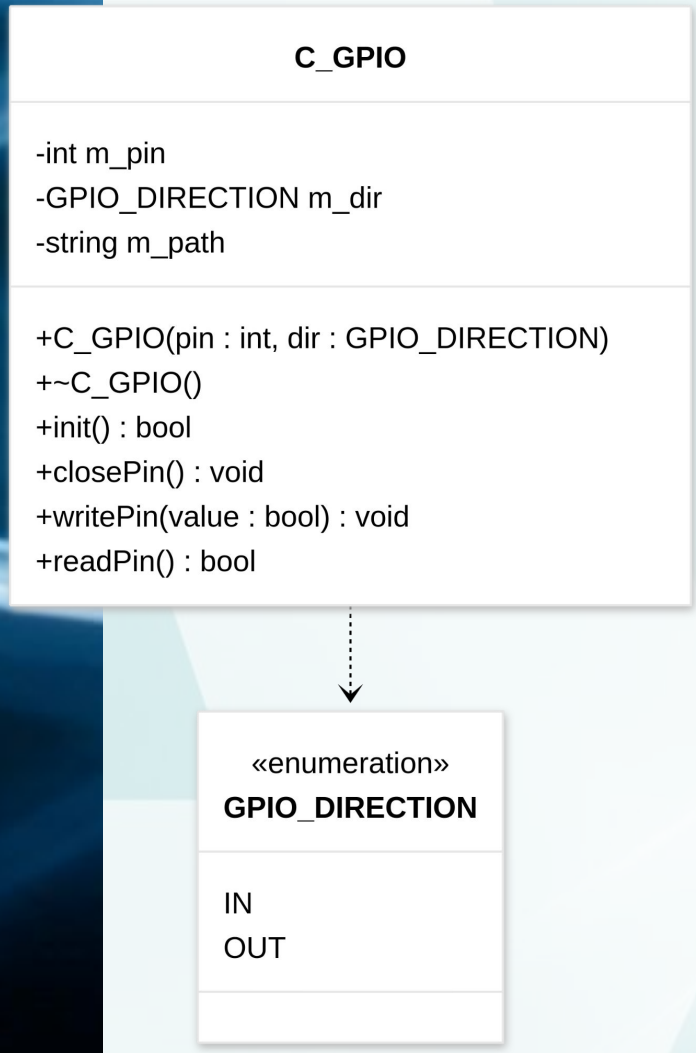
- sysfs for GPIO access

```
# cd /sys/class/gpio
# ls
export       gpiochip512  gpiochip570  unexport
# cd gpiochip512
# ls
base       device     label      ngpio      power      subsystem  uevent
# cat base
512
# cat label
pinctrl-bcm2711
#
```

Linux GPIO controller has a base of 512

```
# cd /sys/class/gpio
# echo 530 > export
# ls
export       gpio530      gpiochip512  gpiochip570  unexport
# cd gpio530
# ls
active_low  device     direction  edge       power      subsystem  uevent     value
#
```

530-512= GPIO Pin 18




```cpp
enum GPIO_DIRECTION { IN, OUT };

class C_GPIO {
public:

    C_GPIO(int pin, GPIO_DIRECTION dir);

    ~C_GPIO();

    //Pin export
    bool init();

    //Pin unexport
    void closePin();

    //output value
    void writePin(bool value);

    //read input value
    bool readPin();

private:
    int m_pin;
    GPIO_DIRECTION m_dir;
    string m_path;//sysfs path
};
```

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```
#define GPIO_BASE 512
```

```
C_GPIO::C_GPIO(int pin, GPIO_DIRECTION dir)
    : m_dir(dir)
{
    m_pin=pin+GPIO_BASE;
    m_path = "/sys/class/gpio/gpio" + to_string(m_pin);
}
```

GPIO Interrupt Device Driver

Creating a device driver that will be sending signals to our main process when there is a change in a specific pin value

ioctl()- used to define the PID to where the signals are sent
write()-used to activate and deactivate interrupts
open()
close()

GPIO


```
#define BCM2708_PERI_BASE       0xfe000000
#define GPIO_BASE (BCM2708_PERI_BASE + 0x200000) // GPIO controller

struct GpioRegisters
{

    uint32_t GPFSEL[6];

    uint32_t ReservedGap[51];

    uint32_t PUP_PDN[4];
};
```

5.2. Register View

The GPIO has the following registers. All accesses are assumed to be 32-bit. The GPIO register base address is 0x7e200000.

Offset	Name	Description
0x00	GPFSEL0	GPIO Function Select 0
0x04	GPFSEL1	GPIO Function Select 1
0x08	GPFSEL2	GPIO Function Select 2
0x0c	GPFSEL3	GPIO Function Select 3
0x10	GPFSEL4	GPIO Function Select 4
0x14	GPFSEL5	GPIO Function Select 5

Module init


```
// 1. Alocar Major/Minor
if ((ret = alloc_chrdev_region(&irqDevice_majorminor, 0, NUM_IRQS, DEVICE_NAME)) < 0) {
    return ret;
}
```

```
// 2. Create class
if (IS_ERR(irqDevice_class = class_create(CLASS_NAME))) {
    unregister_chrdev_region(irqDevice_majorminor, NUM_IRQS);
    return PTR_ERR(irqDevice_class);
}
```

```
// 3.add cdev and associate file operation struct witrh major and each minor number
cdev_init(&c_dev, &irqDevice_fops);
c_dev.owner = THIS_MODULE;

if ((ret = cdev_add(&c_dev, irqDevice_majorminor, NUM_IRQS)) < 0) {
```

```
// 4. Create /dev/irqx files that we can access directly
for (i = 0; i < NUM_IRQS; i++) {
    dev_ret = device_create(irqDevice_class, NULL,
                            MKDEV(MAJOR(irqDevice_majorminor), i),
                            NULL, "irq%d", i);
```

```
// 5.ioremap to make the physical GPIO registers accessible to the kernel
s_pGpioRegisters = (struct GpioRegisters *)ioremap(GPIO_BASE, sizeof(struct GpioRegisters));
```

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```c
struct SensorConfig {
    int gpio_pin;
    unsigned long trigger;
    int signal_id;
    char *name;
    bool enabled;
    unsigned long last_time;
};

static struct SensorConfig my_sensors[] = {
    {23, IRQF_TRIGGER_RISING,  44, "reed_switch", true,0},
    {27, IRQF_TRIGGER_RISING,  45, "pir_sensor",  true,0},
    {22, IRQF_TRIGGER_RISING,  46, "fingerprint", true,0},
    { 5, IRQF_TRIGGER_FALLING, 47, "rfid_reader", true,0}
};
```

```c
// 6. Configure interruption
for (i = 0; i < NUM_IRQS; i++) {
    struct SensorConfig *sensor = &my_sensors[i];
    SetGPIOFunction(s_pGpioRegisters, sensor->gpio_pin, 0);
```

```c
//irq number associated to that gpio pin
gpio_logical = 512 + sensor->gpio_pin;
irq_numbers[i] = gpio_to_irq(gpio_logical);
```

```c
//request irq number to the irq number
ret = request_irq(irq_numbers[i],gpio_irq_handler, sensor->trigger, sensor->name, (void *)sensor);
```

Passing the information from each sensor to the IRQ handler so we can identify which sensor it is

gpio_irq_handler()


```c
static irqreturn_t gpio_irq_handler(int irq, void *dev_id) {

    struct kernel_siginfo info;
    struct SensorConfig *sensor_atual = (struct SensorConfig *)dev_id;

    int pino = sensor_atual->gpio_pin;
    int sinal = sensor_atual->signal_id;


    unsigned long now = jiffies;

    if ((now - sensor_atual->last_time) < msecs_to_jiffies(500)) {
        return IRQ_HANDLED;
    }

    // if the time has passed store the time of new sensor interrupt signal
    sensor_atual->last_time = now;

    if (task != NULL) {
        memset(&info, 0, sizeof(struct kernel_siginfo));
        info.si_signo = sinal;
        info.si_code = SI_QUEUE;
        info.si_int = pino;

        if (send_sig_info(sinal, &info, task) < 0) {
            pr_err("irqModule: Erro ao enviar sinal.\n");
        }
    }
```

Restore sensor specific information

Add debounce to guarantee we don't send a lot of signals in a row

Send signal to our process whose PID Is defined on "task"

System calls


```
// FILE OPERATIONS
static struct file_operations irqDevice_fops = {
    .owner          = THIS_MODULE,
    .open           = irq_device_open,
    .release        = irq_device_close,
    .write          = irq_device_write,     // A
    .read           = NULL,                 // N
    .unlocked_ioctl = irq_device_ioctl,
};
```

```
static long irq_device_ioctl(struct file *file, unsigned int cmd, unsigned long arg) {

    if (cmd == REGIST_PID) {
        //get PID
        task = get_current();
```

```
static int irq_device_open(struct inode* p_inode, struct file *p_file) {

    int minor = iminor(p_inode);

    if (minor >= NUM_IRQS) return -ENODEV;

    // store minor number so we can access to the specific sensor struct in write()
    p_file->private_data = (void *)(unsigned long)minor;
```

```c
static ssize_t irq_device_write(struct file *pfile, const char __user *pbuff, size_t len, loff_t *off) {
    char command;
    unsigned long minor;
    struct SensorConfig *sensor;

    // 1. recover which sensor is(stored in opne())
    minor = (unsigned long)pfile->private_data;
    sensor = &my_sensors[minor]; // Apontar para a configuração correta

    // 2.safely copy user commnad that comes form user space
    if (copy_from_user(&command, pbuff, 1)) return -EFAULT;

    pr_alert("irqModule: Comando '%c' recebido para %s (IRQ %d)\n",
             command, sensor->name, irq_numbers[minor]);

    //3. enable and disable interrupt
    if (command == '0') {
        // Only disable if it is enabled
        if (sensor->enabled == true) {
            disable_irq(irq_numbers[minor]); // disable irq of speific gpio
            sensor->enabled = false;
            pr_alert("irqModule: Interrupção DESATIVADA para %s\n", sensor->name);
        }
    }
    else if (command == '1') {
        // Only enable if it is desabled
        if (sensor->enabled == false) {
            enable_irq(irq_numbers[minor]); //enable irq of specific pin
            sensor->enabled = true;
            pr_alert("irqModule: Interrupção ATIVADA para %s\n", sensor->name);
        }
    }
```

So a peripheral described in this document as being at legacy address 0x7Enn_nnnn is available in the 35-bit addres space at 0x4_7Enn_nnnn, and visible to the ARM at 0x0_FEnn_nnnn if Low Peripheral mode is enabled.

The GPIO has the following registers. All accesses are assumed to be 32-bit. The GPIO register base address is `0x7e200000`.

Offset	Name	Description
0x00	GPFSEL0	GPIO Function Select 0
0x04	GPFSEL1	GPIO Function Select 1
0x08	GPFSEL2	GPIO Function Select 2
0x0c	GPFSEL3	GPIO Function Select 3
0x10	GPFSEL4	GPIO Function Select 4
0x14	GPFSEL5	GPIO Function Select 5
0x1c	GPSET0	GPIO Pin Output Set 0
0x20	GPSET1	GPIO Pin Output Set 1
0x28	GPCLR0	GPIO Pin Output Clear 0
0x2c	GPCLR1	GPIO Pin Output Clear 1

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Shared types

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Defines thread priority

```
// ============================================
// thread priority
// ============================================
enum ThreadPriority_enum : int {
    PRIO_LOW        = 10,
    PRIO_MEDIUM     = 30,
    PRIO_HIGH       = 99
};

// ============================================
// IDs of actuators
// ============================================
enum ActuatorID_enum : uint8_t {
    ID_SERVO_ROOM = 0,
    ID_SERVO_VAULT = 1,
    ID_FAN = 2,
    ID_ALARM_ACTUATOR = 3,
    ID_ACTUATOR_COUNT = 4
};

// ============================================
// IDs of Sensors
// ============================================
enum SensorID_enum : uint8_t {
    ID_SHT31 = 0,
    ID_RDM6300 = 1,
    ID_YRM1001 = 2,
    ID_FINGERPRINT = 3,
    ID_SENSOR_COUNT = 4
};
```

```
// ============================================
// Types of Logs
// ============================================
enum LogType_enum : uint8_t {
    LOG_TYPE_ACTUATOR = 0,
    LOG_TYPE_SENSOR = 1,
    LOG_TYPE_ACCESS = 2,
    LOG_TYPE_SYSTEM = 3,
    LOG_TYPE_ALERT = 4,
    LOG_TYPE_INVENTORY = 5
};

// ============================================
// commands for actuator
// ============================================
struct ActuatorCmd {
    ActuatorID_enum actuatorID;
    uint8_t value;
    // 0 = PWM OFF/LIVRE (Destrancar)
    // > 0 = Ângulo/PWM ON (Trancar)

    ActuatorCmd() : actuatorID(ID_SERVO_ROOM), value(0) {}
    ActuatorCmd(ActuatorID_enum id, uint8_t val)
        : actuatorID(id), value(val) {}
};
```

It categorizes events before they are sent to the database.

commands for the thread tAct

```
struct DatabaseLog {
    LogType_enum logType;
    uint32_t entityID;
    double value;
    double value2;
    uint32_t timestamp;
    char description[266]{};
    DatabaseLog()
        : logType(LOG_TYPE_SYSTEM),
          entityID(0),
          value(0.0),
          value2(0.0),
          timestamp(0) {
        description[0] = '\0';
    }
};
```

Universal event container for logging sensor data, user access, and system alerts

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```
enum e_DbCommand {
    DB_CMD_ENTER_ROOM_RFID,
    DB_CMD_LEAVE_ROOM_RFID,
    DB_CMD_UPDATE_ASSET,
    DB_CMD_USER_IN_PIR,
    DB_CMD_WRITE_LOG,
    DB_CMD_LOGIN,
    DB_CMD_GET_DASHBOARD,
    DB_CMD_GET_SENSORS,
    DB_CMD_GET_ACTUATORS,
    DB_CMD_ADD_USER,
    DB_CMD_DELETE_USER,
    DB_CMD_UPDATE_TEMP_THRESHOLD,
    DB_CMD_UPDATE_SAMPLING_TIME,
    DB_CMD_REGISTER_USER,
    DB_CMD_GET_USERS,
    DB_CMD_CREATE_USER,
    DB_CMD_MODIFY_USER,
    DB_CMD_REMOVE_USER,
    DB_CMD_GET_ASSETS,
    DB_CMD_CREATE_ASSET,
    DB_CMD_MODIFY_ASSET,
    DB_CMD_REMOVE_ASSET,
    DB_CMD_GET_SETTINGS,
    DB_CMD_GET_SETTINGS_THREAD,
    DB_CMD_UPDATE_SETTINGS,
    DB_CMD_FILTER_LOGS,
    DB_CMD_STOP_ENV_SENSOR
};
```

Standardized command set for structured database communication and request handling via IPC

```
struct DatabaseMsg {
    e_DbCommand command;
    union {
        char rfid[11];
        DatabaseLog log;
        Data_RFID_Inventory rfidInventory;
        LoginRequest login;
        UserData user;
        AssetData asset;
        SystemSettings settings;
        LogFilter logFilter;
        uint32_t userId;
    } payload;
};
```

Universal request bridge: all system components to Database

```
struct AuthResponse {
    e_DbCommand command;

    union {
        struct {
            bool authorized;
            uint32_t userId;
            uint32_t accessLevel;
        } auth;

        SystemSettings settings;
    } payload;
};
```

DB-to-Threads: Authentication and sync response

```
struct DbWebResponse {
    bool success;
    char jsonData[8192];
    char errorMsg[256];
};
```

DB-to-Web: JSON data transport for UI.

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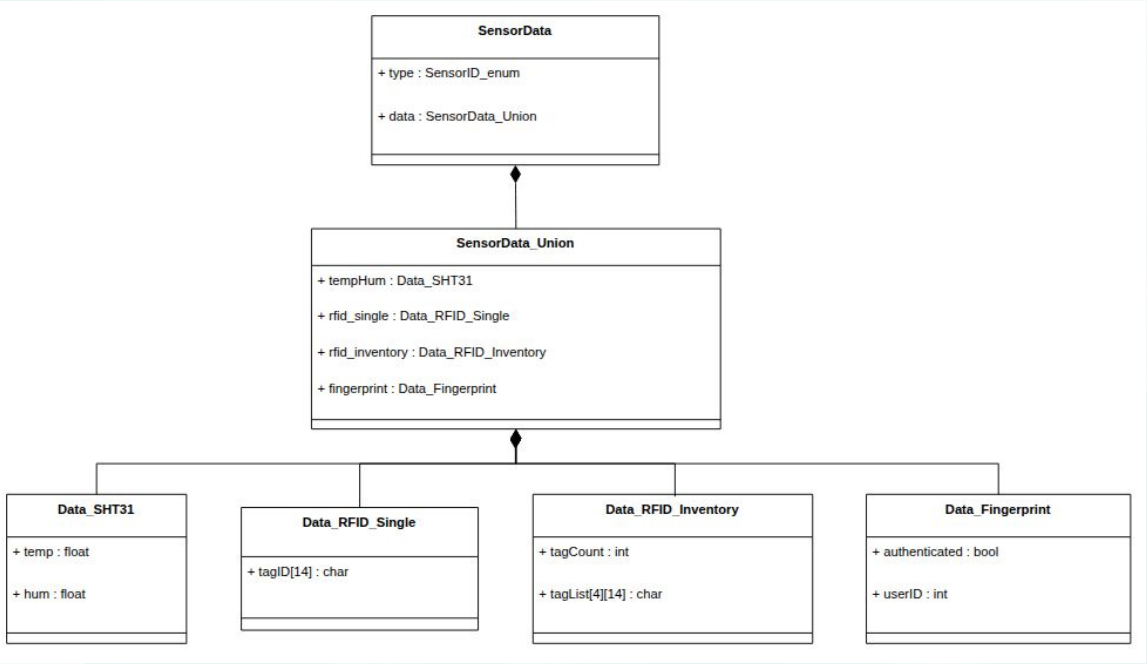
```c
#ifndef C_SENSOR_H
#define C_SENSOR_H
#include "SharedTypes.h"

// Sensor-specific Data Structures
struct Data_SHT31 {
    float temp;
    float hum;
};

struct Data_RFID_Single {
    char tagID[14];
};

struct Data_RFID_Inventory {
    int tagCount;
    char tagList[4][14];
};

struct Data_Fingerprint {
    bool authenticated;
    int userID;
};

union SensorData_Union {
    Data_SHT31 tempHum;
    Data_RFID_Single rfid_single;
    Data_RFID_Inventory rfid_inventory;
    Data_Fingerprint fingerprint;
};

struct SensorData {
    SensorID_enum type;
    SensorData_Union data;
};
```



Class C_Sensor


```
class C_Sensor {
protected:
    SensorID_enum m_sensorID;

public:
    C_Sensor(SensorID_enum id) : m_sensorID(id) {}

    virtual ~C_Sensor();

    virtual bool init() = 0;
    virtual bool read(SensorData* data) = 0;

    SensorID_enum get_ID() const { return m_sensorID; }
};

#endif // C_SENSOR_H
```

Ensures cleanup of derived sensor resources even when using a base pointer

Purely Virtual Methods

C_Sensor
m_sensorID : SensorID_enum
+ C_Sensor(id : SensorID_enum) + virtual ~C_Sensor() *+ virtual init() : bool* *+ virtual read(data : SensorData*) : bool* + get_ID() : : SensorID_enum

Class C_RDM6300

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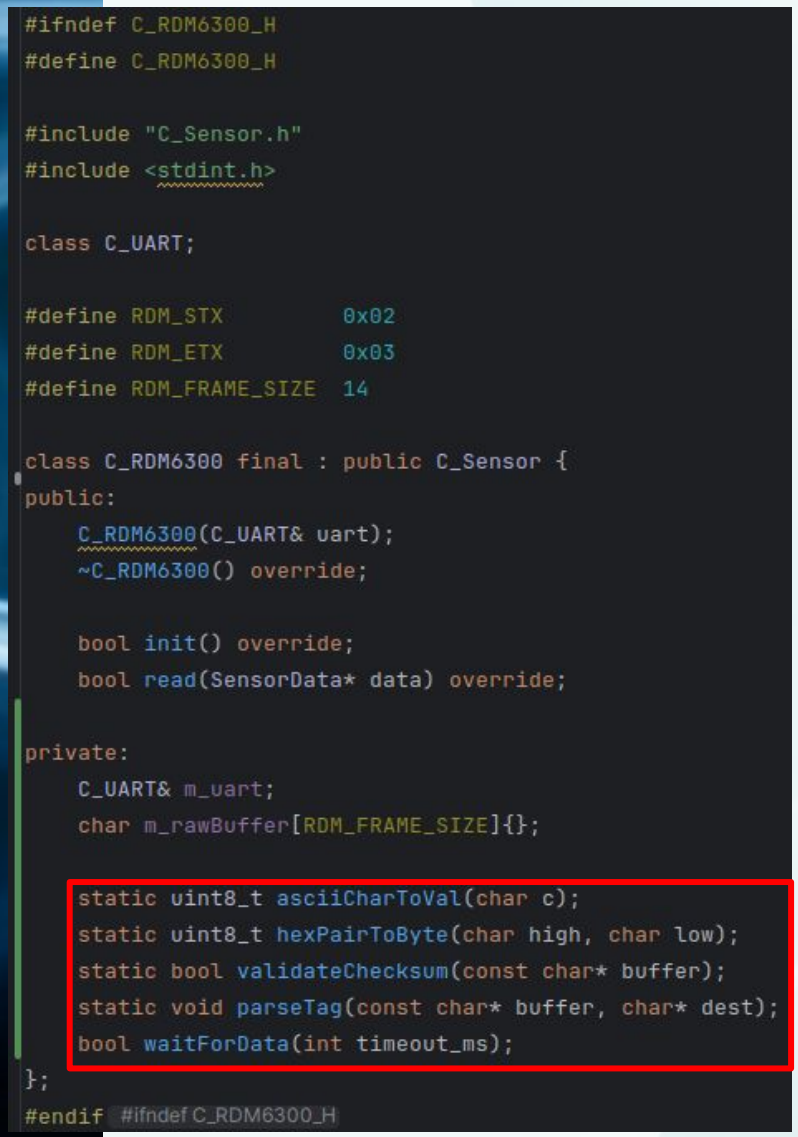
```cpp
#ifndef C_RDM6300_H
#define C_RDM6300_H

#include "C_Sensor.h"
#include <stdint.h>

class C_UART;

#define RDM_STX         0x02
#define RDM_ETX         0x03
#define RDM_FRAME_SIZE  14

class C_RDM6300 final : public C_Sensor {
public:
    C_RDM6300(C_UART& uart);
    ~C_RDM6300() override;

    bool init() override;
    bool read(SensorData* data) override;

private:
    C_UART& m_uart;
    char m_rawBuffer[RDM_FRAME_SIZE]{};

    static uint8_t asciiCharToVal(char c);
    static uint8_t hexPairToByte(char high, char low);
    static bool validateChecksum(const char* buffer);
    static void parseTag(const char* buffer, char* dest);
    bool waitForData(int timeout_ms);
};
#endif #ifndef C_RDM6300_H
```



Internal helpers for protocol validation and data conversion.

```cpp
#include "C_RDM6300.h"
#include "C_UART.h"
#include <iostream>
#include <cstring>
#include <poll.h>
#include <unistd.h>

C_RDM6300::C_RDM6300(C_UART& uart)
    : C_Sensor(ID_RDM6300), m_uart(uart) {}

C_RDM6300::~C_RDM6300() = default;

bool C_RDM6300::init() {
    if (!m_uart.openPort()) return false;
    // RDM6300 default: 9600 baud, 8 data bits, no parity, 1 stop bit
    if (!m_uart.configPort( baud: 9600,  bits: 8,  parity: 'N')) return false;
    return true;
}

// Wait for data with timeout using poll()
// Returns true if data available, false on timeout
bool C_RDM6300::waitForData(int timeout_ms) const {
    struct pollfd pfd;
    pfd.fd = m_uart.getFd();
    pfd.events = POLLIN;     // Wait for read-ready

    // poll() blocks without CPU usage
    int ret = poll(&pfd,  nfds: 1, timeout_ms);

    return (ret > 0);        // >0 = data ready, 0 = timeout, <0 = error
}
```

Configuration of the UART to a baudrate of 9600

This function implements an event-driven architecture using POSIX `poll`. The thread suspends its execution and waits for a kernel notification, waking up strictly when the RFID module transmits data. This mechanism ensures precise timing management for the communication protocol without the need for continuous software checks.

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```
bool C_RDM6300::read(SensorData* data) {
    /* Wait for packet start (STX byte) with 1s timeout.
       If no card present, returns quickly allowing thread exit checks. */
    if (!waitForData( timeout_ms: 1000)) {
        return false;
    }
    // Read STX byte (0x02)
    char header = 0;
    if (m_uart.readBuffer(&header,  len: 1) != 1) return false;
    /* Validate STX - if not 0x02, we have garbage data.
       Clean buffer to avoid synchronization issues. */
    if (header != RDM_STX) {
        char trash[32];
        if (waitForData( timeout_ms: 10)) {    // Brief wait for remaining garbage
            while(m_uart.readBuffer(trash,  len: sizeof(trash)) > 0);
        }
        return false;
    }
    // Start building frame: STX at position 0
    m_rawBuffer[0] = RDM_STX;
    int bytesToRead = 13;    // Remaining: 10 data + 2 checksum + 1 ETX
    int totalRead = 0;
    /* Read frame body with 100ms timeout.
       At 9600 bps (~1ms/byte), 100ms is ample for 13 bytes. */
    while (totalRead < bytesToRead) {
        if (!waitForData( timeout_ms: 100)) {
            return false;    // Frame incomplete/corrupted
        }

        int n = m_uart.readBuffer(m_rawBuffer + 1 + totalRead,
                                  len: bytesToRead - totalRead);
        if (n > 0) {
            totalRead += n;
        } else {
            return false;    // Read error
        }
    }
```

verifies if data was received on UART

Header Validation: Discards random noise to ensure a clean read.

Payload Retrieval: Captures the remaining frame bytes with strict timeout control.

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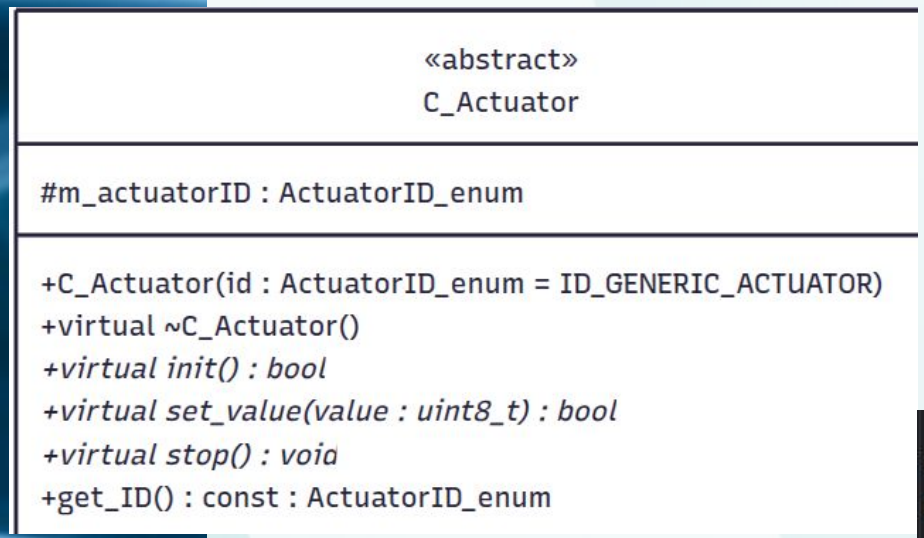
```
    // Validate frame structure and checksum
    bool success = false;
    if (m_rawBuffer[13] == RDM_ETX) {
        if (validateChecksum(m_rawBuffer)) {
            success = true;
            if (data) {
                data->type = ID_RDM6300;
                parseTag(m_rawBuffer, data->data.rfid_single.tagID);
            }
        }
    }
    /* Non-blocking buffer cleanup.
       Remove any leftover bytes to keep UART sync. */
    if (waitForData( timeout_ms: 0)) {
        char trash[64];
        while(m_uart.readBuffer(trash,  len: sizeof(trash)) > 0);
    }
    return success;
}
```

Frame Verification: Validates the End-Byte (ETX) and Checksum before extracting the ID.

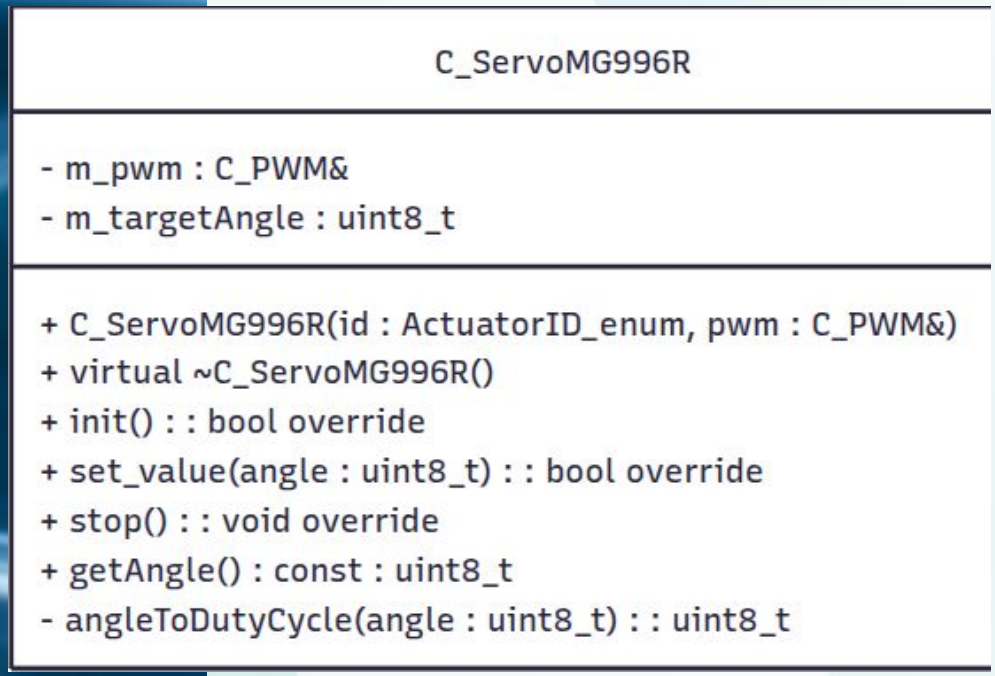
Post-Read Flush: Clears any residual data to prevent overlap in the next cycle.

C_Actuator




```
class C_Actuator {
protected:
    ActuatorID_enum m_actuatorID;

public:
    C_Actuator(ActuatorID_enum id): m_actuatorID(id) {}
    virtual ~C_Actuator();
    virtual bool init() = 0;
    virtual bool set_value(uint8_t value) = 0;
    virtual void stop() = 0;

    ActuatorID_enum get_ID() const { return m_actuatorID; }
};

#endif // C_ACTUATOR_H
```

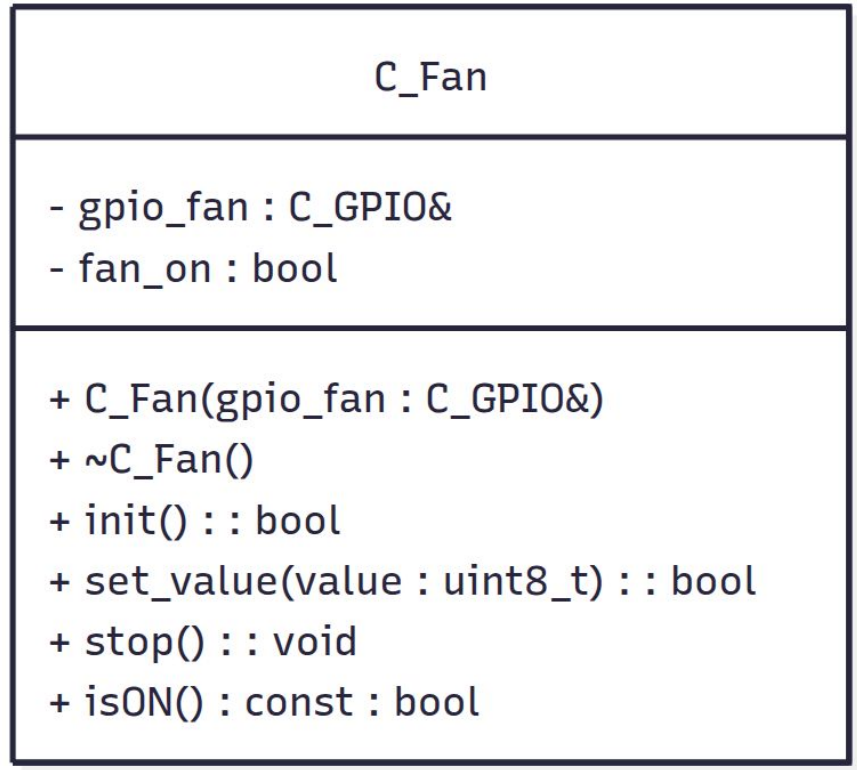
Class C_Servo




```cpp
#ifndef C_SERVOMG996R_H
#define C_SERVOMG996R_H

#include "C_Actuator.h"
#include "../hal/C_PWM.h"
#include <cstdint>

class C_ServoMG996R : public C_Actuator {
public:
    C_ServoMG996R(ActuatorID_enum id, C_PWM& pwm);

    ~C_ServoMG996R() override;

    bool init() override;
    bool set_value(uint8_t angle) override;
    void stop() override;

    uint8_t getAngle() const { return m_targetAngle; }

private:
    C_PWM& m_pwm;
    uint8_t m_targetAngle;

    static uint8_t angleToDutyCycle(uint8_t angle);

};
#endif // C_SERVOMG996R_H
```

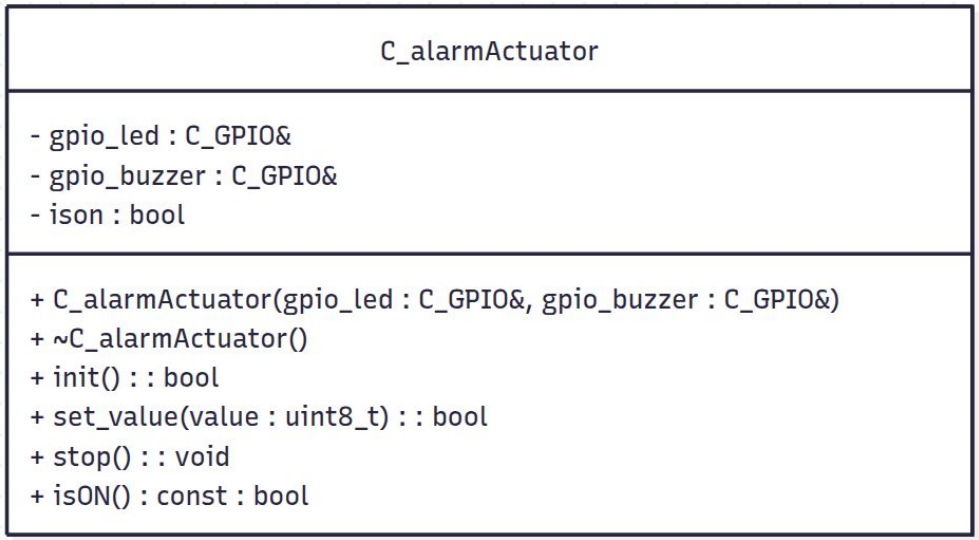
Class C_Fan


```cpp
#ifndef C_FAN_H
#define C_FAN_H

#include "C_Actuator.h"

class C_GPIO;

class C_Fan final: public C_Actuator {

public:
    C_Fan(C_GPIO& gpio_fan);
    ~C_Fan() override;
    bool init() override;
    bool set_value(uint8_t value) override;
    void stop() override;
    bool isON() const {return fan_on;}
private:
    C_GPIO& gpio_fan;
    bool fan_on;
};

#endif #ifndef C_FAN_H
```



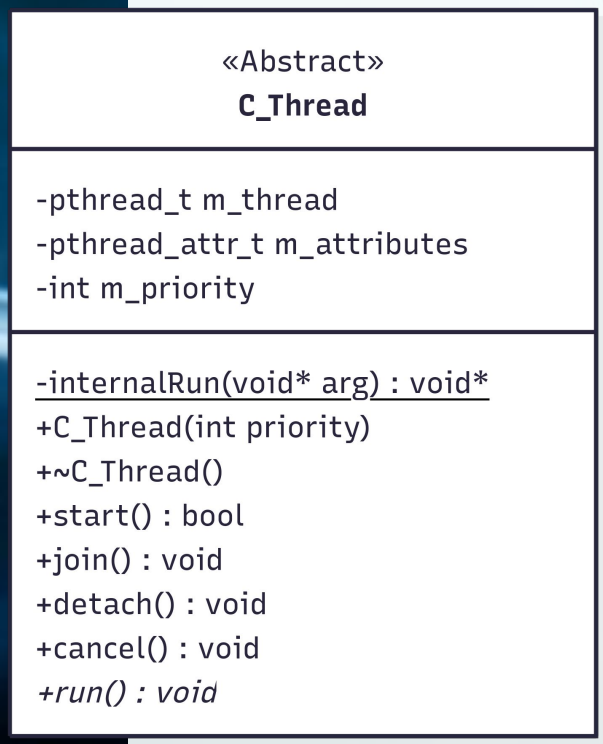
Class C_Alarm


```cpp
#ifndef C_ALARMACTUATOR_H
#define C_ALARMACTUATOR_H
#include "C_Actuator.h"
#include "SharedTypes.h"

class C_GPIO;

class C_alarmActuator final : public C_Actuator {

public:
    C_alarmActuator(C_GPIO& gpio_led, C_GPIO& gpio_buzzer);
    ~C_alarmActuator() override;
    bool init() override;
    bool set_value(uint8_t value) override;
    void stop() override;
    bool isON() const {return ison; }
private:
    C_GPIO& gpio_led;
    C_GPIO& gpio_buzzer;
    bool ison;
};

#endif #ifndef C_ALARMACTUATOR_H
```



Class C_thread

- All of the other thread classes inherit from the C_Thread base class.
- The C_Thread class enables derived classes to avoid pthread library handling




```cpp
class C_Thread {

    // --- Private Data ---
    pthread_t m_thread;           // Stores the Thread ID.
    pthread_attr_t m_attributes;  // Stores settings like priority.
    int m_priority;               // Real-Time priority value.

    // Bridge function required to connect C++ objects to C-style pthreads.
    static void* internalRun(void* arg);

public:

    // Constructor : fill the attributes(priority,...)
    C_Thread(int priority = 0);

    // Destructor: Cleans up the attributes.
    virtual ~C_Thread();

    // --- Execution ---

    // Calls 'pthread_create'(threadid, attributs and run function)
    bool start();

    // Wrapper: Calls 'pthread_join'
    void join();

    // Wrapper: Calls 'pthread_detach'
    void detach();

    // Wrapper: Calls 'pthread_cancel'
    void cancel();


    // Pure Virtual Function. Derived classes must implement their own run!
    virtual void run() = 0;
};
```

```
bool C_Thread::start() {

    int result = pthread_create(&m_thread, &m_attributes, internalRun, arg: this);
```

Pointer to the object that called start function

```
function
extern int pthread_create(
    pthread_t *__newthread,
    const pthread_attr_t *__attr,
    void *(*__start_routine)(void *),
    void *__arg) noexcept(true)
```

We want to use pthread_create for a specif object and with this start the specific run function of that object

Can't do it directly because pthread_create is not expecting a c++ type method that has an implicit "this" pointer to the specific object

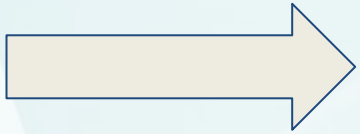
```
static void* internalRun(void* arg);
```

```
void* C_Thread::internalRun(void* arg) {
    C_Thread* threadObj = (C_Thread*)arg;//casting back to C_thread object pointer

    if (threadObj != NULL) {
        threadObj->run();
    }

    return NULL;
}
```

Class C_Monitor

Condition Variable + Mutex

C_Monitor
-pthread_mutex_t m_mutex -pthread_cond_t m_cond
+C_Monitor() +~C_Monitor() +wait() +signal()



```
class C_Monitor {
    pthread_mutex_t m_mutex;
    pthread_cond_t m_cond;

public:
    C_Monitor();
    ~C_Monitor();

    void wait();
    void signal();
};
```

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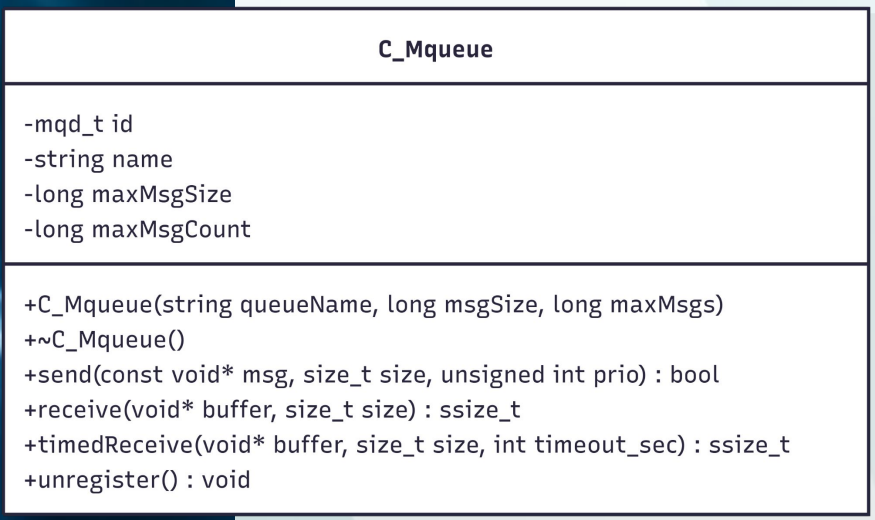
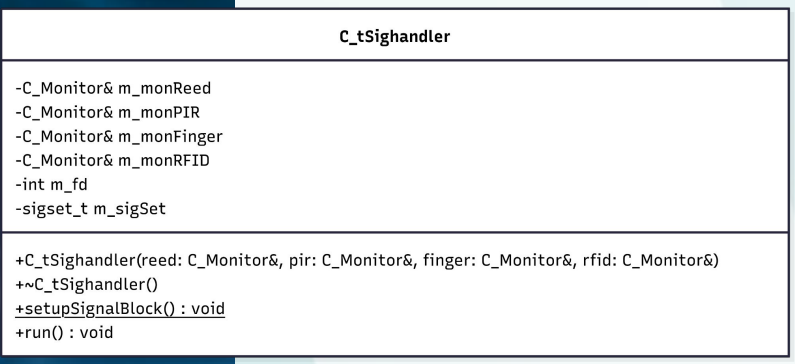
```
C_Monitor::C_Monitor() {
    if (pthread_mutex_init(&m_mutex, mutexattr: NULL) != 0){
        cerr << "Mutex init failed" << endl;
    }
    if (pthread_cond_init(&m_cond, cond_attr: NULL) != 0) {
        cerr << "Cond init failed" << endl;
    }
}

C_Monitor::~C_Monitor() {
    pthread_mutex_destroy(&m_mutex);
    pthread_cond_destroy(&m_cond);
}

void C_Monitor::wait() {

    pthread_mutex_lock(&m_mutex);
    pthread_cond_wait(&m_cond, &m_mutex);
    pthread_mutex_unlock(&m_mutex);
}


void C_Monitor::signal() {
    pthread_mutex_lock(&m_mutex);
    pthread_cond_signal(&m_cond);
    pthread_mutex_unlock(&m_mutex);
```

Class C_mqueue




```
class C_Mqueue {
private:
    mqd_t id;
    string name;
    long maxMsgSize;//message nax byte syze
    long maxMsgCount;//max message number inside queue

public:
    C_Mqueue(string queueName, long msgSize = 1024, long maxMsgs = 10);

    ~C_Mqueue();


    bool send(const void* msg, size_t size, unsigned int prio = 0);
    //thread waits for message queue
    ssize_t receive(void* buffer, size_t size);
    //thread waits for a specific time and if there is no message just keeps going
    ssize_t timedReceive(void* buffer, size_t size, int timeout_sec);
    void unregister();
};

#endif // C_MQUEUE_H
```

Class C_tSighandler




```
class C_tSighandler : public C_Thread {

    C_Monitor& m_monReed;
    C_Monitor& m_monPIR;
    C_Monitor& m_monFinger;
    C_Monitor& m_monRFID;

    int m_fd;
    sigset_t m_sigSet;

public:
    C_tSighandler(C_Monitor& reed, C_Monitor& pir, C_Monitor& finger, C_Monitor& rfid);

    virtual ~C_tSighandler() override;

    // should be the first thing to be called in the main
    static void setupSignalBlock();


    void run() override;
};
```

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```
void C_tSighandler::setupSignalBlock() {

    sigset_t set;
    sigemptyset(&set);
    sigaddset(&set, signo:43);
    sigaddset(&set, signo:44);
    sigaddset(&set, signo:45);
    sigaddset(&set, signo:46);
    sigaddset(&set, signo:47);
    sigaddset(&set, signo:48);

    if (pthread_sigmask( how: SIG_BLOCK, &set, oldmask: NULL) != 0) {
        perror( s: "Erro ao bloquear sinais");
    }
}

void C_tSighandler::run() {
    siginfo_t info;


    m_fd = open( file: "/dev/irq0", oflag: O_WRONLY);
    if (m_fd < 0 || ioctl(m_fd, request: REGIST_PID, 0) < 0) {
        std::cerr << "[Sighandler] ERRO: Não foi possivel conectar a
        return;
    }

    std::cout << "[Sighandler] Pronto. À espera de eventos de hardwa
```

```
int sig = sigtimedwait(&m_sigSet, &info, &timeout);

if (sig < 0) {
    if (errno == EAGAIN) {
        continue;
    }
    continue;
}

int pino = info.si_int;

switch (sig) {
    case 43:
        std::cout << "[Hardware] Reed Switch detetado
        m_monReed_vault.signal();
        break;

    case 44:
        std::cout << "[Hardware] Reed Switch detetado
        m_monReed_room.signal();
        break;
    case 45:
        std::cout << "[Hardware] Movimento PIR detetad
        m_monPIR.signal();
        break;
    case 46:
        std::cout << "[Hardware] Digital lida no pino
        m_monFinger.signal();
        break;
    case 47:
        std::cout << "[Hardware] RFID entrada no pino
        m_monRFID_entry.signal();
        break;
    case 48:
        std::cout << "[Hardware] RFID saida aproximad
        m_monRFID_exit.signal();
}
```

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C_tVerifyRoomAccess

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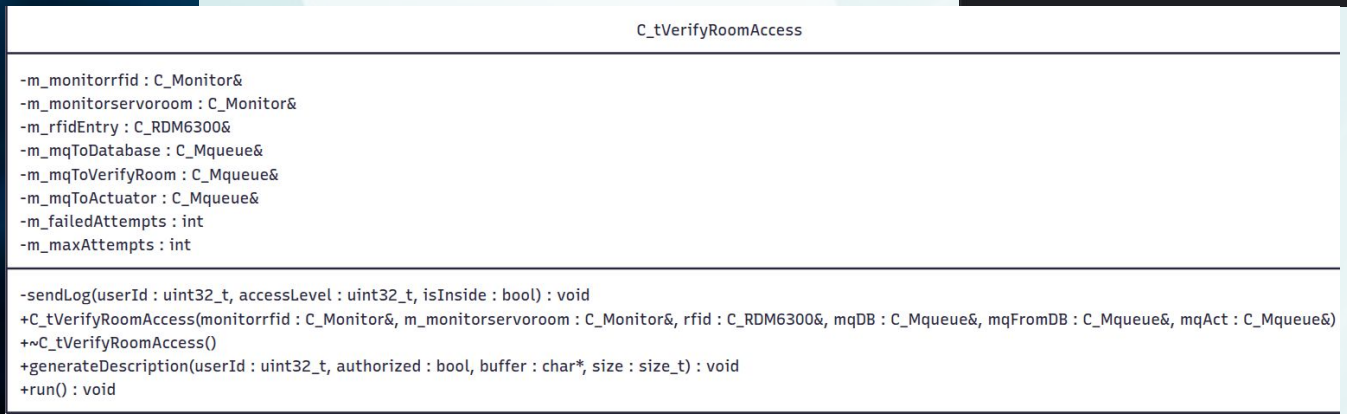
```
class C_tVerifyRoomAccess : public C_Thread {
private:

    C_Monitor& m_monitorrfid;
    C_Monitor& m_monitorservoroom;
    C_RDM6300& m_rfidEntry;
    C_Mqueue& m_mqToDatabase;
    C_Mqueue& m_mqToVerifyRoom;
    C_Mqueue& m_mqToActuator;

    int m_failedAttempts;
    int m_maxAttempts;

    void sendLog(uint32_t userId, uint32_t accessLevel, bool isInside);

public:

    _tVerifyRoomAccess(C_Monitor& monitorrfid,
                       C_Monitor& m_monitorservoroom,
                       C_Mqueue& mqDB,
                       C_Mqueue& mqFromDB,
                       C_Mqueue& mqAct);

    virtual ~C_tVerifyRoomAccess();
    void generateDescription(uint32_t userId, bool authorized, char* buffer, size_t size);
    void run() override;
};

#endif  #ifndef C_TVERIFYROOMACCESS_H
```



C_tVerifyRoomAccess


```
void C_tVerifyRoomAccess::run() {
    std::cout << "[VerifyRoomAccess] Thread iniciada. À espera de tags..." << std::endl;


    while (!stopRequested()) {

        if (m_monitorrfid.timedWait( seconds: 1)) {
            continue;
        }

        SensorData data = {};
        if (m_rfidEntry.read(&data)) {
            const char* rfidRead = data.data.rfid_single.tagID;
            std::cout << "[RFID entry] Cartão lido: " << rfidRead << std::endl;

            DatabaseMsg msg = {};
            msg.command = DB_CMD_ENTER_ROOM_RFID;
            strncpy(msg.payload.rfid, rfidRead, n: sizeof(msg.payload.rfid) - 1);
            msg.payload.rfid[sizeof(msg.payload.rfid) - 1] = '\0';
            m_mqToDatabase.send(&msg, size: sizeof(msg));
            std::cout << "m enviafa: " << std::endl;
```

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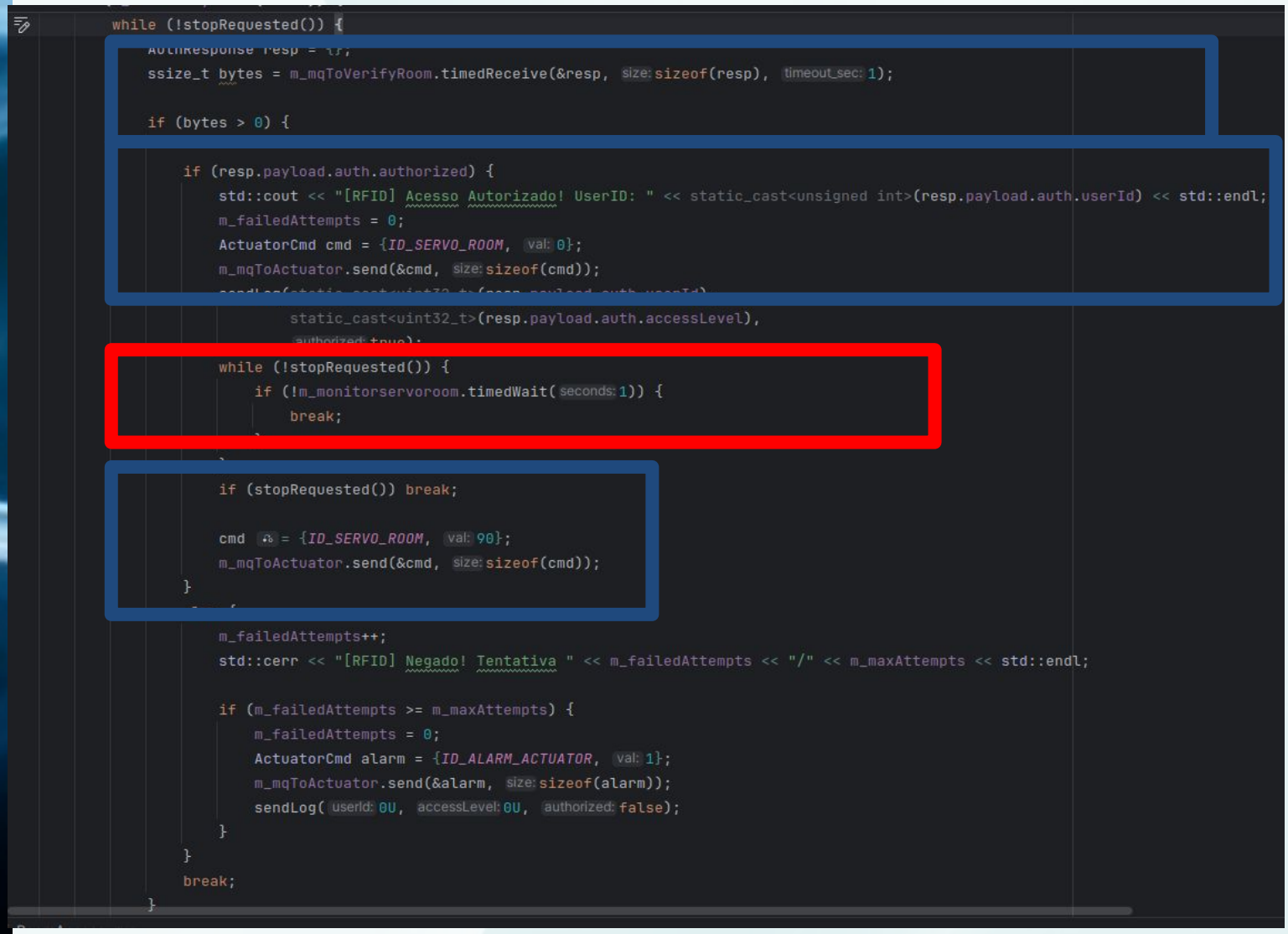

Class dDatabase


```
class dDatabase {
public:
    dDatabase(const std::string& dbPath,
              C_Mqueue& mqDb,
              C_Mqueue& mqRfidIn,
              C_Mqueue& mqRfidOut,
              C_Mqueue& mqFinger,
              C_Mqueue& mqCheckMovement,
              C_Mqueue& mqToWeb,
              C_Mqueue& mqToEnv);
    ~dDatabase();

    bool open();
    void close();
    bool initializeSchema();
    void processDbMessage(const DatabaseMsg& msg);

private:
    sqlite3* m_db;
    std::string m_dbPath;

    C_Mqueue& m_mqToDatabase;
    C_Mqueue& m_mqToVerifyRoom;
    C_Mqueue& m_mqToLeaveRoom;
    C_Mqueue& m_mqToFingerprint;
    C_Mqueue& m_mqToCheckMovement;
    C_Mqueue& m_mqToWeb;
    C_Mqueue& m_mqToEnvThread;
```

```
void handleAccessRequest(const char* rfid, bool isEntering);
void handleScanInventory(const Data_RFID_Inventory& inventory);
void handleCheckUserInPir();
void handleLogin(const LoginRequest& login);
void handleGetDashboard();
void handleGetSensors();
void handleGetActuators();
void handleInsertLog(const DatabaseLog& log);
void updateSensorTable(uint8_t entityID, double value, double value2, uint32_t timestamp)
void updateActuatorTable(uint8_t entityID, double value, uint32_t timestamp);

void handleRegisterUser(const UserData& user);
void handleGetUsers();
void handleCreateUser(const UserData& user);
void handleModifyUser(const UserData& user);
void handleRemoveUser(uint32_t userId);

void handleGetAssets();
void handleCreateAsset(const AssetData& asset);
void handleModifyAsset(const AssetData& asset);
void handleRemoveAsset(const char* tag);

void handleGetSettings();
void handleGetSettingsForThread();

void handleUpdateSettings(const SystemSettings& settings);

void handleFilterLogs(const LogFilter& filter);
```

database open()


```
}

bool dDatabase::open() {
    int result = sqlite3_open( filename: m_dbPath.c_str(), &m_db);

    if (result != SQLITE_OK) {
        std::cerr << "ERRO: Não foi possível abrir a base de dados: "
                  << sqlite3_errmsg(m_db) << std::endl;
        return false;
    }

    std::array<unsigned char, kDbKeyLength> key = buildDbKey();

    if (sqlite3_key(m_db,  pKey: key.data(), static_cast<int>(key.size())) != SQLITE_OK) {
        std::cerr << "ERRO: Falha ao definir chave SQLCipher: "
                  << sqlite3_errmsg(m_db) << std::endl;
        secureZero(key.data(),  len: key.size());
        sqlite3_close(m_db);
        m_db = nullptr;
        return false;
    }

    secureZero(key.data(),  len: key.size());

    std::cout << "SUCESSO: Ligado ao ficheiro: " << m_dbPath << std::endl;
    return true;
}
```

```
constexpr std::array<uint8_t, kDbKeyLength> kDbKeyPart = {
    0x31, 0x76, 0x9a, 0x40, 0x7b, 0xa3, 0x1d, 0xc4,
    0x2a, 0x8f, 0x61, 0x05, 0xde, 0x19, 0x73, 0xb0,
    0x4c, 0x8a, 0x16, 0xf2, 0x9c, 0x5b, 0x07, 0xd8,
    0x21, 0x90, 0x4e, 0xbc, 0x63, 0x0a, 0xe1, 0x57
};

constexpr std::array<uint8_t, kDbKeyLength> kDbKeyMask = {
    0x8d, 0x1f, 0x27, 0xbb, 0x19, 0x5e, 0x88, 0x22,
    0x9c, 0x30, 0xf5, 0x6a, 0x4b, 0xac, 0x02, 0x1e,
    0x77, 0x0d, 0xf9, 0x45, 0x6b, 0x92, 0xfe, 0x14,
    0xc8, 0x59, 0x73, 0x02, 0xbd, 0xa4, 0x38, 0xc1
};

std::array<unsigned char, kDbKeyLength> buildDbKey() {
    std::array<unsigned char, kDbKeyLength> key{};
    for (size_t i = 0; i < key.size(); ++i) {
        key[i] = static_cast<unsigned char>(kDbKeyPart[i] ^ kDbKeyMask[i]);
    }
    return key;
}

void secureZero(void* data, size_t len) {
    volatile unsigned char* p = static_cast<volatile unsigned char*>(data);
    while (len--) {
        *p++ = 0;
    }
}
```

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database initializeSchema()

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Database Tables
Users
Logs
Assets
Sensors
Actuators
SystemSettings

sqlite> PRAGMA table_info(Users);

cid	name	type	notnull	dflt_value	pk
0	UserID	INTEGER	0		1
1	Name	TEXT	0		0
2	RFID_Card	TEXT	0		0
3	FingerprintID	INTEGER	0		0
4	Password	TEXT	0		0
5	AccessLevel	INTEGER	0		0
6	IsInside	INTEGER	0	0	0

sqlite> PRAGMA table_info(Logs);

cid	name	type	notnull	dflt_value	pk
0	LogsID	INTEGER	0		1
1	EntityID	INTEGER	0		0
2	Timestamp	INTEGER	0		0
3	LogType	INTEGER	0		0
4	Description	TEXT	0		0
5	Value	INTEGER	0		0
6	Value2	INTEGER	0	0	0

sqlite> PRAGMA table_info(Assets);

cid	name	type	notnull	dflt_value	pk
0	AssetID	INTEGER	0		1
1	Name	TEXT	0		0
2	RFID_Tag	TEXT	0		0
3	State	TEXT	0		0
4	LastRead	INTEGER	0		0

sqlite> PRAGMA table_info(Sensors);

cid	name	type	notnull	dflt_value	pk
0	SensorID	INTEGER	0		1
1	Type	TEXT	0		0
2	Value	INTEGER	0		0
3	LastUpdate	INTEGER	0	0	0

sqlite> PRAGMA table_info(Actuators);

cid	name	type	notnull	dflt_value	pk
0	ActuatorID	INTEGER	0		1
1	Type	TEXT	0		0
2	State	INTEGER	0		0
3	LastUpdate	INTEGER	0	0	0

sqlite> PRAGMA table_info(SystemSettings);

cid	name	type	notnull	dflt_value	pk
0	ID	INTEGER	0		1
1	TempThreshold	INTEGER	0	30	0
2	SamplingTime	INTEGER	0	600	0

Database main


```
while (!g_stop) {
    DatabaseMsg msg{};
    ssize_t bytesRead = mqToDb.timedReceive(&msg, size: sizeof(DatabaseMsg), timeout_sec: 1);
    if (bytesRead > 0) {
        db.processDbMessage(msg);
    }
}
```

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```
void dDatabase::processDbMessage(const DatabaseMsg &msg) {
        case DB_CMD_ENTER_ROOM_RFID:
            handleAccessRequest(msg.payload.rfid, isEntering: true);
        case DB_CMD_LEAVE_ROOM_RFID:
            handleAccessRequest(msg.payload.rfid, isEntering: false);
            break;
        case DB_CMD_UPDATE_ASSET:
            handleScanInventory(msg.payload.rfidInventory);
            break;
        case DB_CMD_WRITE_LOG:
            handleInsertLog(msg.payload.log);
            break;
        case DB_CMD_USER_IN_PIR:
            handleCheckUserInPir();
            break;
        case DB_CMD_LOGIN:
            handleLogin(msg.payload.login);
            break;
        case DB_CMD_GET_DASHBOARD:
            handleGetDashboard();
            break;
        case DB_CMD_GET_SENSORS:
            handleGetSensors();
            break;
        case DB_CMD_GET_ACTUATORS:
            handleGetActuators();
            break;
        case DB_CMD_REGISTER_USER:
            handleRegisterUser(msg.payload.user);
            break;
        case DB_CMD_GET_USERS:
            handleGetUsers();
            break;
        case DB_CMD_CREATE_USER:
            handleCreateUser(msg.payload.user);
```

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```
void dDatabase::handleAccessRequest(const char* rfid, bool isEntering) {
    sqlite3_stmt* stmt;

    AuthResponse resp = {};
    resp.command = isEntering ? DB_CMD_ENTER_ROOM_RFID : DB_CMD_LEAVE_ROOM_RFID;

    const char* sqlSelect = "SELECT UserID, AccessLevel FROM Users WHERE RFID_Card = ?;";
    if (sqlite3_prepare_v2(m_db, sqlSelect, nByte: -1, &stmt, pzTail: nullptr) == SQLITE_OK) {
        sqlite3_bind_text(stmt, 1, rfid, -1, SQLITE_TRANSIENT);
        if (sqlite3_step(stmt) == SQLITE_ROW) {
            resp.payload.auth.authorized = true;
            resp.payload.auth.userId = static_cast<uint32_t>(sqlite3_column_int(stmt, iCol: 0));
            resp.payload.auth.accessLevel = static_cast<uint32_t>(sqlite3_column_int(stmt, iCol: 1));

            cout << "\n[CHECK] User Encontrado!" << endl;
            cout << " > UserID:      " << resp.payload.auth.userId << endl;
            cout << " > AccessLevel: " << resp.payload.auth.accessLevel << endl;
            cout << "----------------------------" << endl;
        }
        sqlite3_finalize(stmt);
    }
    if (resp.payload.auth.authorized) {
        int newState = isEntering ? 1 : 0;

        const char* sqlUpdate = "UPDATE Users SET IsInside = ? WHERE UserID = ?;";
        if (sqlite3_prepare_v2(m_db, sqlUpdate, nByte: -1, &stmt, pzTail: nullptr) == SQLITE_OK) {
            sqlite3_bind_int(stmt, 1, newState);
            sqlite3_bind_int(stmt, 2, static_cast<int>(resp.payload.auth.userId));
            sqlite3_step(stmt);
            sqlite3_finalize(stmt);
        }
    }

    if (isEntering) {
        m_mqToVerifyRoom.send(&resp, size: sizeof(resp));
    } else {
        m_mqToLeaveRoom.send(&resp, size: sizeof(resp));
    }
}
```

Class Webserver


```cpp
#ifndef DWEBSERVER_H
#define DWEBSERVER_H

#include <string>
#include <map>
#include <mutex>
#include "mongoose.h"
#include "nlohmann/json.hpp"
#include "C_Mqueue.h"
#include "SharedTypes.h"

struct SessionData {
    uint32_t userId;
    std::string username;
    uint32_t accessLevel;  // 0=Viewer, 1=Room, 2=Room+Vault
    time_t expires;
};

class dWebServer {
private:
    struct mg_mgr m_mgr;
    C_Mqueue& m_mqToDatabase;
    C_Mqueue& m_mqFromDatabase;
    std::map<std::string, SessionData> m_sessions;
    std::mutex m_sessionMutex;
    int m_port;
    bool m_running;

public:
    dWebServer(C_Mqueue& toDb, C_Mqueue& fromDb, int port = 8080);
    ~dWebServer();

    bool start();
    void stop();
    void run();
```

```cpp
private:
    static void eventHandler(struct mg_connection* c, int ev, void* ev_data);

    void handleLogin(struct mg_connection* c, struct mg_http_message* hm);
    void handleRegister(struct mg_connection* c, struct mg_http_message* hm);
    void handleLogout(struct mg_connection* c, struct mg_http_message* hm);

    void handleDashboard(struct mg_connection* c, struct mg_http_message* hm);
    void handleSensors(struct mg_connection* c, struct mg_http_message* hm);
    void handleActuators(struct mg_connection* c, struct mg_http_message* hm);
    void handleLogsFilter(struct mg_connection* c, struct mg_http_message* hm);

    void handleUsers(struct mg_connection* c, struct mg_http_message* hm);
    void handleUsersById(struct mg_connection* c, struct mg_http_message* hm);
    void handleAssets(struct mg_connection* c, struct mg_http_message* hm);
    void handleAssetsById(struct mg_connection* c, struct mg_http_message* hm);
    void handleSettings(struct mg_connection* c, struct mg_http_message* hm);

    std::string generateToken();
    bool validateSession(struct mg_http_message* hm, SessionData& outSession);
    void cleanExpiredSessions();

    void sendJson(struct mg_connection* c, int statusCode, const nlohmann::json& data);
    void sendError(struct mg_connection* c, int statusCode, const std::string& message);

    static bool matchUri(const struct mg_str* uri, const char* pattern);
    static bool matchPrefix(const struct mg_str* uri, const char* pattern);
};

#endif #ifndef DWEBSERVER_H
```

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```cpp
bool dWebServer::start() {
    std::string addr = "http://10.42.0.163:" + std::to_string(m_port) + "/";

    if (mg_http_listen(&m_mgr, url: addr.c_str(), eventHandler, fn_data: this) == nullptr) {
        std::cerr << "[WebServer] Failed to start on port " << m_port << std::endl;
        return false;
    }

    m_running = true;
    std::cout << "[WebServer] Listening on " << addr << std::endl;
    return true;
}
```

Service Initialization: Binding the HTTP listener to the network interface and attaching the global event dispatcher.

```cpp
void dWebServer::stop() {
    m_running = false;
}
```

```cpp
void dWebServer::run() {
    while (m_running) {
        mg_mgr_poll(&m_mgr, ms: 1000);
        cleanExpiredSessions();
    }
}
```

Main Execution Loop: Orchestrating non-blocking event processing and background session maintenance.

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```
void dWebServer::eventHandler(struct mg_connection* c, int ev, void* ev_data) {
    if (ev == MG_EV_HTTP_MSG) {
        struct mg_http_message* hm = (struct mg_http_message*)ev_data;
        dWebServer* self = (dWebServer*)c->fn_data;

        if (matchUri(&hm->uri, pattern: "/api/login")) {
            self->handleLogin(c, hm);
        }
        else if (matchUri(&hm->uri, pattern: "/api/register")) {
            self->handleRegister(c, hm);
        }
        else if (matchUri(&hm->uri, pattern: "/api/logout")) {
            self->handleLogout(c, hm);
        }
        else if (matchUri(&hm->uri, pattern: "/api/dashboard")) {
            self->handleDashboard(c, hm);
        }
        else if (matchUri(&hm->uri, pattern: "/api/sensors")) {
            self->handleSensors(c, hm);
        }
        else if (matchUri(&hm->uri, pattern: "/api/actuators")) {
            self->handleActuators(c, hm);
        }
        else if (matchUri(&hm->uri, pattern: "/api/logs/filter")) {
            self->handleLogsFilter(c, hm);
        }
        else if (matchPrefix(&hm->uri, pattern: "/api/assets/")) {
            self->handleAssetsById(c, hm);
        }
        else if (matchPrefix(&hm->uri, pattern: "/api/users/")) {
            self->handleUsersById(c, hm);
        }
        else if (matchUri(&hm->uri, pattern: "/api/users")) {
            self->handleUsers(c, hm);
        }
        else if (matchUri(&hm->uri, pattern: "/api/assets")) {
            self->handleAssets(c, hm);
        }
```

```
        else if (matchUri(&hm->uri, pattern: "/api/settings")) {
            self->handleSettings(c, hm);
        }
        else if (matchUri(&hm->uri, pattern: "/")) {
            mg_http_reply(c, status_code: 302, headers: "Location: /login.html\r\n", body_fmt: "");
        }
        else {
            SessionData session;
            bool logado = self->validateSession(hm, [&] session);
            if (matchUri(&hm->uri, pattern: "/users.html") ||
                matchUri(&hm->uri, pattern: "/settings.html") ||
                matchUri(&hm->uri, pattern: "/assets.html")) {
                if (!logado || session.accessLevel < 1) {
                    mg_http_reply(c, status_code: 302, headers: "Location: /login.html\r\n", body_fmt: "");
                    return;
                }
            }
            struct mg_http_serve_opts opts;
            memset(&opts, c: 0, n: sizeof(opts));
            opts.root_dir = "/root/SecureAsset/web";
            mg_http_serve_dir(c, hm, &opts);
        }
    }
}
```

Dispatches incoming HTTP requests to specialized backend logic based on URI matching

Static Content Delivery: Automatically serves HTML, CSS, and JS files from the server's root directory.

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```
void dWebServer::handleSensors(struct mg_connection* c, struct mg_http_message* hm) {
    SessionData session;
    if (!validateSession(hm, [&]session)) {
        sendError(c, statusCode: 401, message: "Not authenticated");
        return;
    }

    DatabaseMsg msg = {};
    msg.command = DB_CMD_GET_SENSORS;
    m_mqToDatabase.send(&msg, size: sizeof(msg));

    DbWebResponse resp = {};
    ssize_t bytes = m_mqFromDatabase.timedReceive(&resp, size: sizeof(resp), timeout_sec: 2);

    if (bytes > 0 && resp.success) {
        sendJson(c, statusCode: 200, data: nlohmann::json::parse([&]resp.jsonData));
    } else {
        sendError(c, statusCode: 500, message: "Failed to get sensors");
    }
}
```

Verifies the user session and fetches current sensor data from the database to send back as JSON.

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Web Interface

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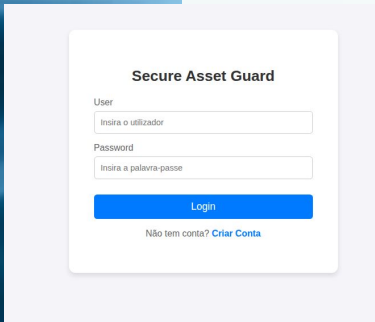
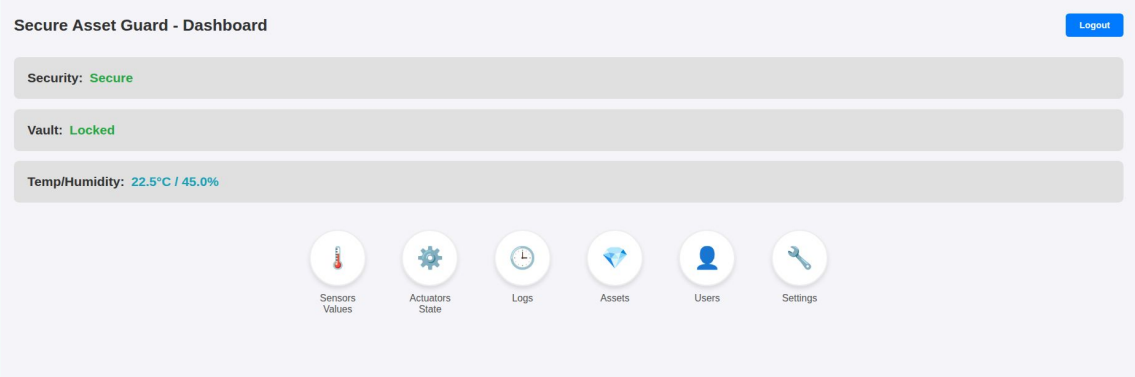
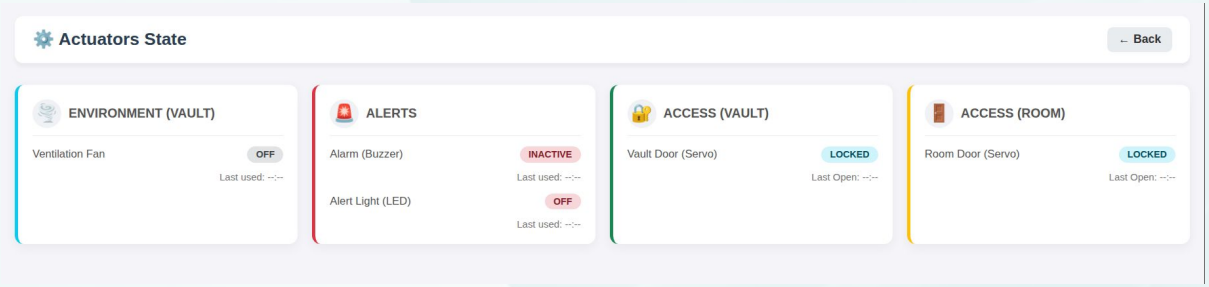
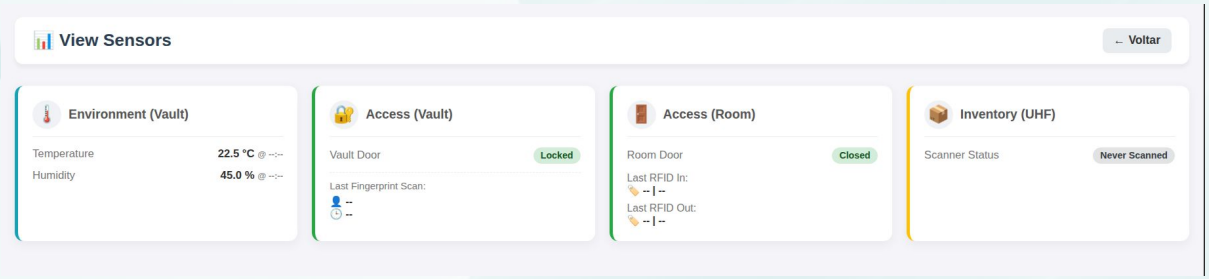
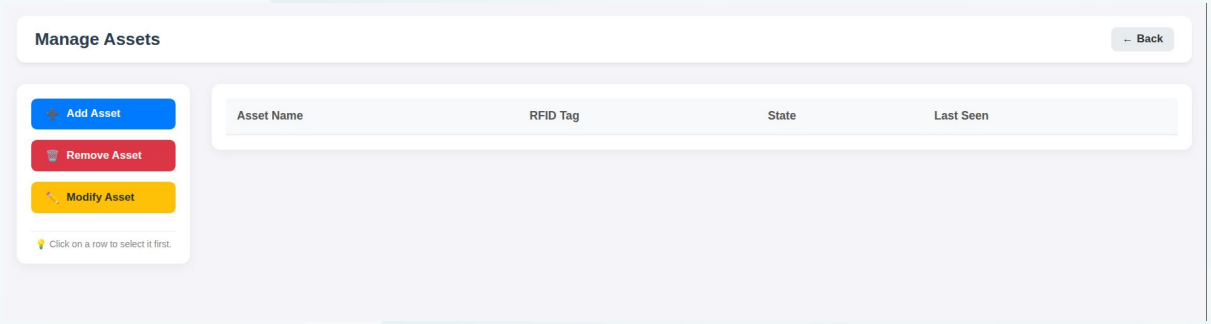
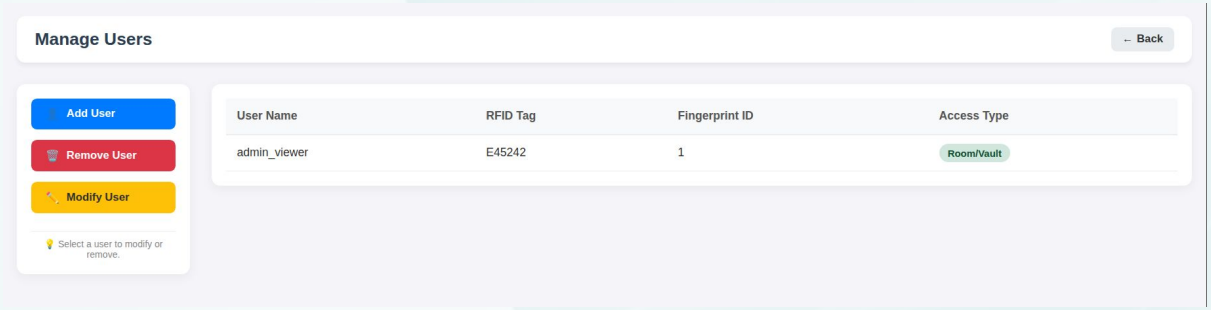


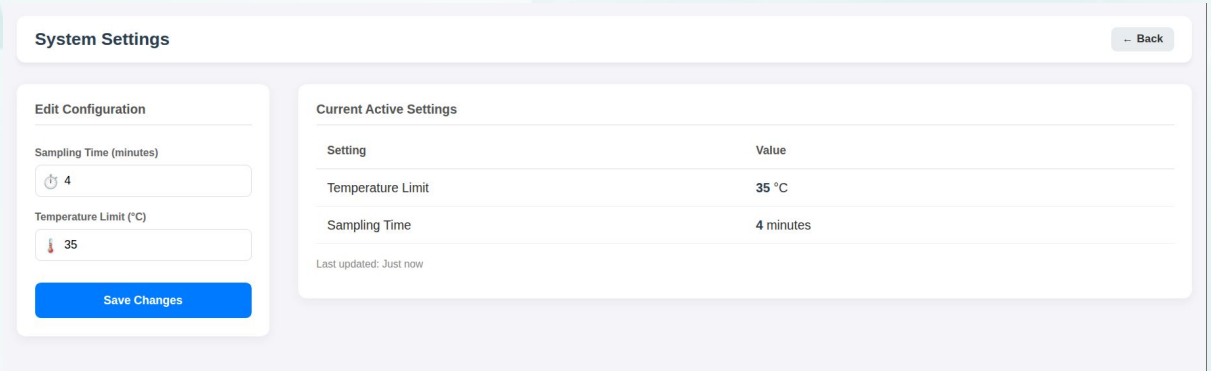




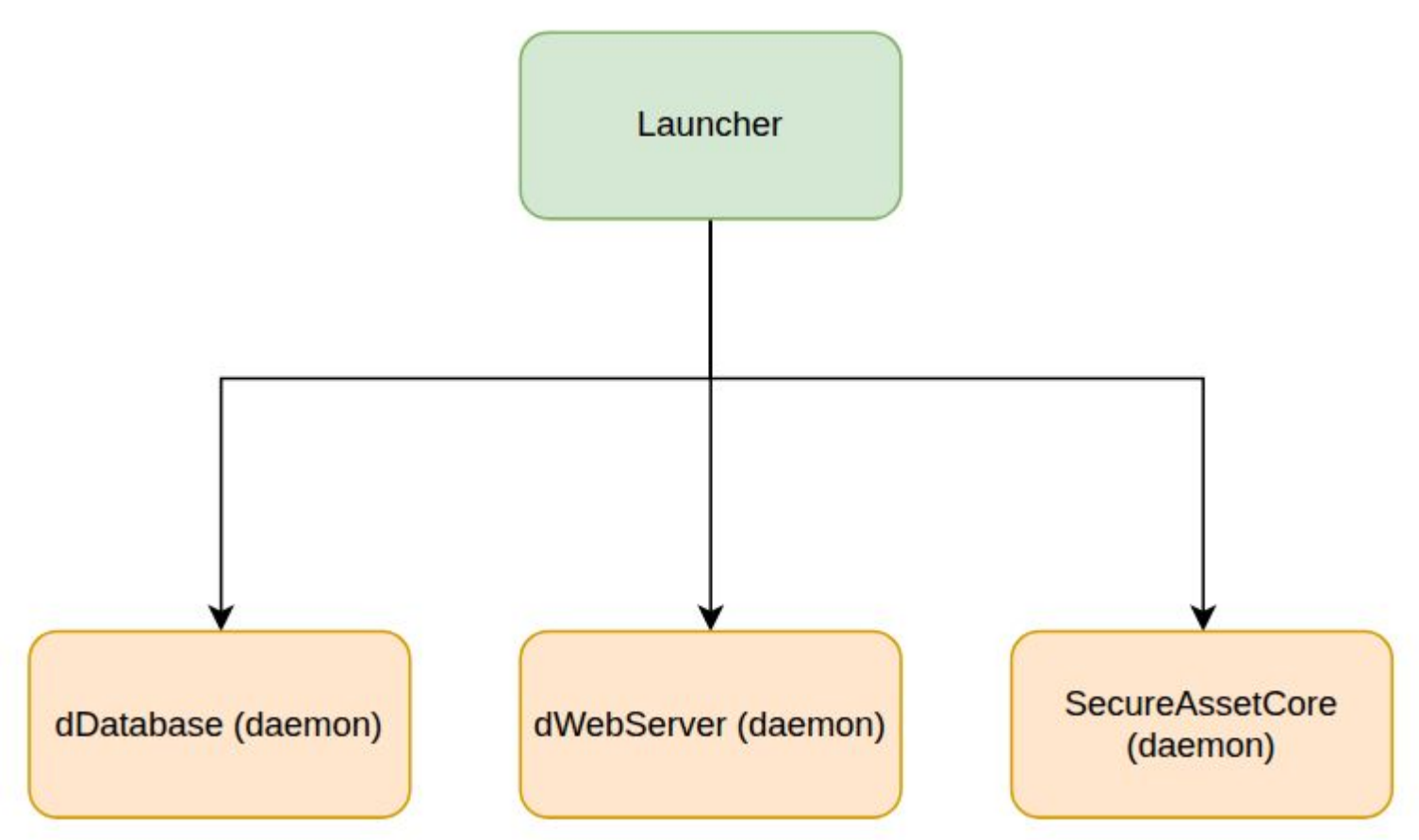



Manage Assets
← Back
Add Asset
Remove Asset
Modify Asset
Click on a row to select it first.
Asset Name
RFID Tag
State
Last Seen


Manage Users
← Back
Add User
Remove User
Modify User
Select a user to modify or remove.
User Name
RFID Tag
Fingerprint ID
Access Type
admin_viewer
E45242
1
Room/Vault


System Settings
← Back
Edit Configuration
Sampling Time (minutes)
4
Temperature Limit (°C)
35
Save Changes
Current Active Settings
Setting
Value
Temperature Limit
35 °C
Sampling Time
4 minutes
Last updated: Just now

Startup & Shutdown Daemonization



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Startup & Shutdown Daemonization

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```
// Create message queues (launcher is the owner)
std::vector<std::unique_ptr<C_Mqueue>> mqs;
try {
    mqs.push_back( x: std::make_unique<C_Mqueue>( queueName: "/mq_to_db",  msgSize: sizeof(DatabaseMsg),  maxMsgs: 20,  createNew: true));
    mqs.push_back( x: std::make_unique<C_Mqueue>( queueName: "/mq_to_actuator",  msgSize: sizeof(ActuatorCmd),  maxMsgs: 20,  createNew: true));
    mqs.push_back( x: std::make_unique<C_Mqueue>( queueName: "/mq_rfid_in",  msgSize: sizeof(AuthResponse),  maxMsgs: 10,  createNew: true));
    mqs.push_back( x: std::make_unique<C_Mqueue>( queueName: "/mq_rfid_out",  msgSize: sizeof(AuthResponse),  maxMsgs: 10,  createNew: true));
    mqs.push_back( x: std::make_unique<C_Mqueue>( queueName: "/mq_move",  msgSize: sizeof(AuthResponse),  maxMsgs: 10,  createNew: true));
    mqs.push_back( x: std::make_unique<C_Mqueue>( queueName: "/mq_finger",  msgSize: sizeof(AuthResponse),  maxMsgs: 10,  createNew: true));
    mqs.push_back( x: std::make_unique<C_Mqueue>( queueName: "/mq_db_to_env",  msgSize: sizeof(AuthResponse),  maxMsgs: 10,  createNew: true));
    mqs.push_back( x: std::make_unique<C_Mqueue>( queueName: "/mq_db_to_web",  msgSize: sizeof(DbWebResponse),  maxMsgs: 10,  createNew: true));
    std::cout << "[Wrapper] Message queues created successfully.\n";
} catch (const std::exception& e) {
    std::cerr << "[Wrapper] ERROR creating queues: " << e.what() << std::endl;
    return EXIT_FAILURE;
}

std::string execDir = getExecutableDir();

DaemonInfo db = launchDaemon( binName:  "dDatabase",  binPath: execDir + "/dDatabase");
if (db.pid < 0) return EXIT_FAILURE;
std::cout << "[Wrapper] dDatabase PID " << db.pid << std::endl;

DaemonInfo web = launchDaemon( binName:  "dWebServer",  binPath: execDir + "/dWebServer");
if (web.pid < 0) {
    stopDaemon( [&] db);
    return EXIT_FAILURE;
}
std::cout << "[Wrapper] dWebServer PID " << web.pid << std::endl;

DaemonInfo core = launchDaemon( binName:  "SecureAssetCore",  binPath: execDir + "/SecureAssetCore");
if (core.pid < 0) {
    stopDaemon( [&] web);
    stopDaemon( [&] db);
    return EXIT_FAILURE;
}
```

Wrapper initializes all IPC message queues and launches each daemon using an explicit READY handshake

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```
static DaemonInfo launchDaemon(const std::string& binName, const std::string& binPath)
    DaemonInfo info;
    info.name = binName;
    int sv_startup[2];
    int sv_shutdown[2];
    if (socketpair( domain: AF_UNIX,  type: SOCK_STREAM,  protocol: 0, sv_startup) < 0) {
        std::perror( s: "socketpair startup");
        return info;
    }
    if (socketpair( domain: AF_UNIX,  type: SOCK_STREAM,  protocol: 0, sv_shutdown) < 0) {
        std::perror( s: "socketpair shutdown");
        close(sv_startup[0]);
        close(sv_startup[1]);
        return info;
    }
```

Create two dedicated IPC channels: STARTUP (READY) and SHUTDOWN (ACK), eliminating timing guesses

Event-driven startup: wrapper blocks on poll(timeout) and proceeds only after receiving READY + PID

```
// Daemon ends must survive exec()
if (!clearCloExec(sv_startup[1]) || !clearCloExec(sv_shutdown[1])) {
    std::perror( s: "fcntl FD_CLOEXEC");
    close(sv_startup[0]); close(sv_startup[1]);
    close(sv_shutdown[0]); close(sv_shutdown[1]);
    return info;
}
char fdStartupStr[16], fdShutdownStr[16];
std::snprintf(fdStartupStr,  maxlen: sizeof(fdStartupStr),  format: "%d", sv_startup[1]);
std::snprintf(fdShutdownStr,  maxlen: sizeof(fdShutdownStr),  format: "%d", sv_shutdown[1]);
setenv( name: "NOTIFY_FD",  value: fdStartupStr,  replace: 1);
setenv( name: "SHUTDOWN_FD",  value: fdShutdownStr,  replace: 1);
pid_t child = fork();
if (child < 0) {
    std::perror( s: "fork");
    close(sv_startup[0]); close(sv_startup[1]);
    close(sv_shutdown[0]); close(sv_shutdown[1]);
    unsetenv( name: "NOTIFY_FD");
    unsetenv( name: "SHUTDOWN_FD");
    return info;
}
```

Pass the handshake file descriptors through exec() using environment variables (FD_CLOEXEC cleared), then spawn the daemon process.

```
waitpid( pid: child,  stat_loc: nullptr,  options: 0);
// Wait PID from daemon (startup handshake), timeout 5s
pollfd pfd{};
pfd.fd = sv_startup[0];
pfd.events = POLLIN | POLLHUP;
int ret = poll(&pfd,  nfds: 1,  timeout: 5000);
if (ret > 0 && (pfd.revents & (POLLIN | POLLHUP))) {
    info.pid = readPidFromFd(sv_startup[0]);
} else {
    std::cerr << "[Wrapper] ERROR: startup timeout for " << binName << std::endl;
}
close(sv_startup[0]);
if (info.pid <= 0) {
    // startup failed: close shutdown channel too
    close(sv_shutdown[0]);
    info.shutdown_fd = -1;
    info.pid = -1;
    return info;
}
// Keep shutdown read end for later
info.shutdown_fd = sv_shutdown[0];
return info;
```

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```cpp
static void daemonize(const char* pidfile, const char* logfile = nullptr) {
    pid_t pid = fork();
    if (pid < 0) std::exit( status: EXIT_FAILURE);
    if (pid > 0) std::exit( status: EXIT_SUCCESS);

    if (setsid() < 0) std::exit( status: EXIT_FAILURE);

    std::signal( sig: SIGHUP, SIG_IGN);

    pid = fork();
    if (pid < 0) std::exit( status: EXIT_FAILURE);
    if (pid > 0) std::exit( status: EXIT_SUCCESS);

    umask( mask: 0);
    chdir( path: "/");

    close( fd: STDIN_FILENO);
    open( file: "/dev/null",  oflag: O_RDONLY);

    if (logfile) {
        int fd = open(logfile,  oflag: O_WRONLY | O_CREAT | O_APPEND, 0644);
        if (fd >= 0) {
            dup2(fd,  fd2: STDOUT_FILENO);
            dup2(fd,  fd2: STDERR_FILENO);
            close(fd);
        } else {
            close( fd: STDOUT_FILENO); close( fd: STDERR_FILENO);
            open( file: "/dev/null",  oflag: O_WRONLY);
            open( file: "/dev/null",  oflag: O_WRONLY);
        }
    } else {
        close( fd: STDOUT_FILENO); close( fd: STDERR_FILENO);
        open( file: "/dev/null",  oflag: O_WRONLY);
        open( file: "/dev/null",  oflag: O_WRONLY);
    }

    int fd = open(pidfile,  oflag: O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (fd >= 0) {
        char buf[32];
        ssize_t n = std::snprintf(buf,  maxlen: sizeof(buf),  format: "%d\n", getpid());
        (void)write(fd, buf, n);
        close(fd);
    }
}
```

Uses a double-fork pattern to detach the service from the terminal and create an independent session

Normalize daemon environment: reset umask and move working directory to /

Redirects output to log files and records the Process ID (PID) for system management.

Startup & Shutdown Daemonization


```
static void stopDaemon(DaemonInfo& daemon) {
    if (daemon.pid <= 0) return;

    std::cout << "[Wrapper] Stopping " << daemon.name << " (PID " << daemon.pid << ")..." << std::endl;

    // Request graceful termination
    if (kill(daemon.pid, sig: SIGTERM) != 0 && errno == ESRCH) {
        if (daemon.shutdown_fd >= 0) { close(daemon.shutdown_fd); daemon.shutdown_fd = -1; }
        return;
    }

    if (daemon.shutdown_fd >= 0) {
        pollfd pfd{};
        pfd.fd = daemon.shutdown_fd;
        pfd.events = POLLIN | POLLHUP;

        int ret = poll(&pfd, nfds: 1, timeout: 5000);
        if (ret > 0 && (pfd.revents & (POLLIN | POLLHUP))) {
            // Accept either explicit "OK" or just EOF
            char buf[16];
            (void)read(daemon.shutdown_fd, buf, nbytes: sizeof(buf));
            close(daemon.shutdown_fd);
            daemon.shutdown_fd = -1;
            std::cout << "[Wrapper] " << daemon.name << " stopped (ACK/EOF)." << std::endl;
            return;
        }

        // Timeout or error
        close(daemon.shutdown_fd);
        daemon.shutdown_fd = -1;
    }

    std::cout << "[Wrapper] " << daemon.name << " did not respond. Forcing SIGKILL." << std::endl;
    kill(daemon.pid, sig: SIGKILL);
}
```

Graceful shutdown request: send SIGTERM to let the daemon clean up

Wait for shutdown acknowledgement with a bounded timeout (poll).

Startup & Shutdown Daemonization


```
static void sendShutdownAck() {
    if (g_shutdown_fd >= 0) {
        const char* ack = "OK\n";
        (void)write(g_shutdown_fd, ack, n:3);
        close(g_shutdown_fd);
        g_shutdown_fd = -1;
    }
}
```

On termination, the daemon confirms cleanup completion by sending an 'OK' acknowledgement and closing the channel

```
// Startup notify (PID or -1)
if (notify_fd >= 0) {
    char buf[64];
    int len = ok ? std::snprintf(buf, maxlen: sizeof(buf), format: "%d\n", getpid())
                 : std::snprintf(buf, maxlen: sizeof(buf), format: "-1\n");
    (void)write(notify_fd, buf, len);
    close(notify_fd);
    notify_fd = -1;
}
```

After initialization succeeds, the daemon notifies the wrapper with its PID (or -1 on failure)